PHYSICS

CURRENT ELECTRICITY_MEGA_XII



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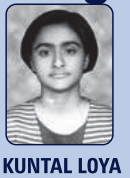
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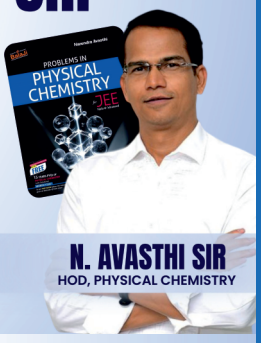
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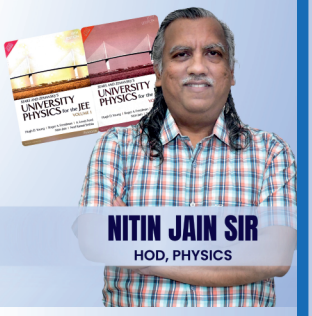


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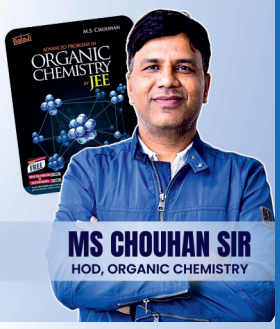


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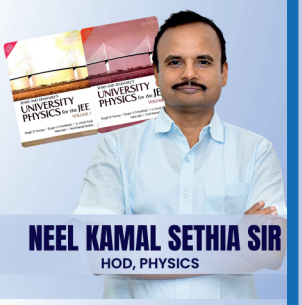


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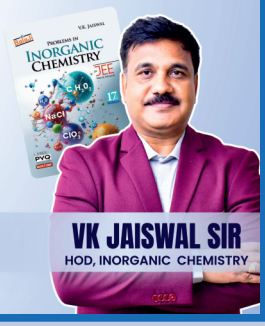


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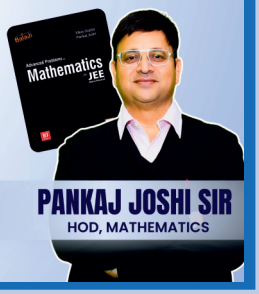


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PHYSICS

TARGET IIT JEE

MEGA-XII

CURRENT ELECTRICITY

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CURRENT ELECTRICITY

1. ELECTRIC CURRENT :

Electric charges in motion constitute an electric current. Any medium having practically free electric charges, free to migrate is a conductor of electricity. The electric charge flows from higher potential energy state to lower potential energy state. Positive charge flows from higher to lower potential and negative charge flows from lower to higher. Metals such as gold, silver, copper, aluminium etc. are good conductors.

2. ELECTRIC CURRENT IN A CONDUCTOR :

In absence of potential difference across a conductor no net current flows through a corss section. When a potential difference is applied across a conductor the charge carriers (electrons in case of metallic conductors) flow in a definite direction which constitutes a net current in it. These electrons are not accelerated by electric field in the conductor produced by potential difference across the conductor. They move with a constant drift velocity. The direction of current is along the flow of positive charge (or opposite to flow of negative charge). $i = nv_d eA$, where V_d = drift velocity.

3. CHARGE AND CURRENT :

The strength of the current *i* is the rate at which the electric charges are flowing. If a charge Q coulomb passes through a given cross section of the conductor in t second the current I through the conductor is given by $I = \frac{Q}{t} = \frac{\text{Coulomb}}{\text{second}} = \frac{Q}{t}$ ampere .

Ampere is the unit of current . If *i* is not constant then $i = \frac{dq}{dt}$, where dq is net charge transported at a section in time dt.

In a current carrying conductor we can define a vector which gives the direction as current per unit normal, cross sectional area.

Thus $\vec{J} = \frac{I}{S}\hat{n}$ or $I = \vec{J} \cdot \vec{S}$

Where $\hat{n}$ is the unit vector in the direction of the flow of current. For random J or S, we use $I = \int \vec{J} \cdot \vec{ds}$

4. RELATION IN J, E AND ν_d :

In conductors drift vol. of electrons is proportional to the electric field in side the conductor as – $\nu_d = \mu E$

where μ is the mobility of electrons current density is given as $J = \frac{I}{A} = ne\nu_d = ne(\mu E) = \sigma E$; where $\sigma = ne\mu$ is called conductivity of material and we can also write $\rho = \frac{1}{\sigma} \rightarrow$ resistivity of material.

Thus $E = \rho J$. It is called as differential form of Ohm's Law.

5. SOURCES OF POTENTIAL DIFFERENCE & ELECTROMOTIVE FORCE :

Dry cells , secondary cells , generator and thermo couple are the devices used for producing potential difference in an electric circuit. The potential difference between the two terminals of a source when no energy is drawn from it is called the **" Electromotive force"** or **" EMF "** of the source. The unit of potential difference is volt.

1 volt = 1 Amphere × 1 Ohm.

6. ELECTRICAL RESISTANCE :

The property of a substance which opposes the flow of electric current through it is termed as electrical resistance. Electrical resistance depends on the size, geometery, temperature and internal structure of the conductor.

7. LAW OF RESISTANCE :

The resistance R offered by a conductor depends on the following factors :

$R \alpha L$ (length of the conductor) ;

$R \alpha \frac{l}{A}$ (cross section area of the conductor)

at a given temperature $R = \rho \frac{l}{A}$.

Where ρ is the resistivity of the material of the conductor at the given temperature. It is also known as **specific resistance** of the material .

8. DEPENDENCE OF RESISTANCE ON TEMPERATURE :

The resistance of most conductors and all pure metals increases with temperature , but there are a few in which resistance decreases with temperature . If R_o & R be the resistance of a conductor at 0º C and θº C, then it is found that $R = R_o (1 + \alpha \theta)$.

Here we assume that the dimensions of resistance does not change with temperature if expansion coefficient of material is considerable.

Then instead of resistance we use same property for resistivity as $\rho = \rho_0 (1 + \alpha\theta)$

The materials for which resistance decreases with temperature, the temperature coefficient of resistance is negative.

Where α is called the temperature co-efficient of resistance. The unit of α is K^{-1} of $°C^{-1}$ reciprocal of resistivity is called conductivity and reciprocal of resistance is called conductance (G). S.I. unit of G is ohm.

9. OHM'S LAW :

Ohm's law is the most fundamental of all the laws in electricity . It says that the current through the cross section or the conductor is proportional to the applied potential difference under the given physical condition. V = R I . Ohm's law is applicable to only metalic conductors .

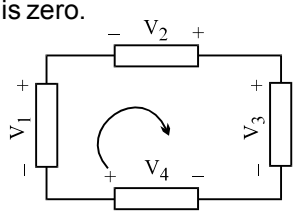
10. KRICHHOFF'S LAW'S :

I - Law (Junction law or Nodal Analysis) : This law is based on law of conservation of charge . It states that " The algebric sum of the currents meeting at a point is zero " or total currents entering a junction equals total current leaving the junction.

$\Sigma I_{in} = \Sigma I_{out}$. It is also known as KCL (Kirchhoff's current law) .

II - Law (Loop analysis) : The algebric sum of all the voltages in closed circuit is zero.

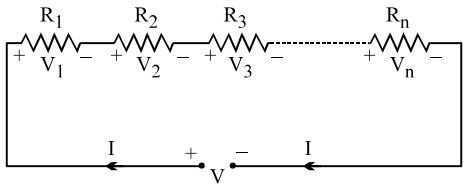
$\Sigma IR + \Sigma EMF = 0$ in a closed loop. The closed loop can be traversed in any direction. While traversing a loop if higher potential point is entered, put a + ve sign in expression or if lower potential point is entered put a negative sign.



$-V_1 - V_2 + V_3 - V_4 = 0$. Boxes may contain resistor or battery or any other element (linear or non-linear).

It is also known as **KVL (Kirchhoff's voltage law)** .

11. COMBINATION OF RESISTANCES :



A number of resistances can be connected and all the complecated combinations can be reduced to two different types, namely series and parallel.

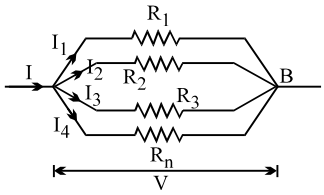
(i) Resistance In Series :

When the resistances are connected end toend then they are said to be in series . The current through each resistor is same . The effective resistance appearing across the battery .

$R = R_1 + R_2 + R_3 + ... + R_n$ and

$V = V_1 + V_2 + V_3 + ... + V_n$.

The voltage across a resistor is proportional to the resistance



$$V_1 = \frac{R_1}{R_1+R_2+.....+R_n}\ V; V_2 = \frac{R_2}{R_1+R_2+.....+R_n}\ V;\ \text{etc}$$

(ii) Resistance In Parallel :

A parallel circuit of resistors is one in which the same voltage is applied across all the components in a parallel grouping of resistors R_1, R_2, R_3, ..., R_n .

Conclusions :

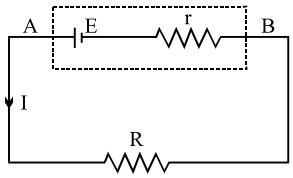
(a) Potential difference across each resistor is same .

(b) $I = I_1 + I_2 + I_3 + ... I_n$.

(c) Effective resistance (R) then $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}+.....+\frac{1}{R_n}$.

(d) Current in different resistors is inversally proportional to the resistance .

$$I_1 : I_2 : ..... I_n = \frac{1}{R_1}:\frac{1}{R_2}:\frac{1}{R_3}:.....:\frac{1}{R_n}\ .$$

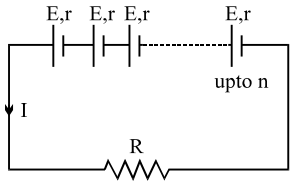


$$I_1 = \frac{G_1}{G_1+G_2+.....+G_n}\ I\ ,\ I_2 = \frac{G_2}{G_1+G_2+.....+G_n}\ I,\ \text{etc}\ .$$

where $G = \frac{I}{R}$ = Conductance of a resistor.

12. EMF OF A CELL & ITS INTERNAL RESISTANCE :

If a cell of emf E and internal resistance r be connected with a resistance R the total resistance of the circuit is (R + r) .



$I = \frac{E}{R+r}$; $V_{AB} = \frac{E}{R+r}$ where

E = Terminal voltage of the battery .If $r \to 0$, cell is Ideal & $V \to E$.

13. GROUPING OF CELLS :

(i) Cells In Series :

Let there be n cells each of emf E , arranged in series.Let r be the internal resistance of each cell.

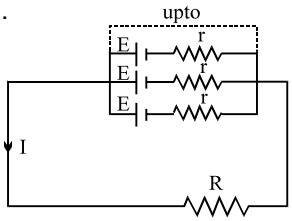
The total emf = n E. Current in the circuit $I = \frac{nE}{R+nr}$.

If nr << R then $I = \frac{nE}{R}$ $\longrightarrow$ Series combination should be used .

If nr >> R then $I = \frac{E}{r}$ $\longrightarrow$ Series combination should not be used .

(ii) Cells In Parallel :

If m cells each of emf E & internal resistance r be connected in parallel and if this combination be connected to an external resistance then the emf of the circuit = E .



Internal resistance of the circuit = $\frac{r}{m}$.

$$I = \frac{E}{R+\frac{r}{m}} = \frac{mE}{mR+r} .$$

If mR << r ; $I = \frac{mE}{r} \longrightarrow$ Parallel combination should be used .

If mR >> r ; $I = \frac{E}{R} \longrightarrow$ Parallel combination should not be used .

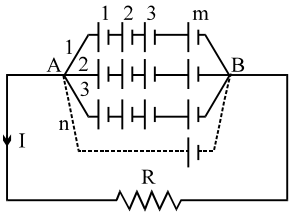
(iii) Cells In Multiple Arc :

mn = number of identical cells .

n = number of rows

m = number of cells in each rows .

The combination of cells is equivalent to single cell of :



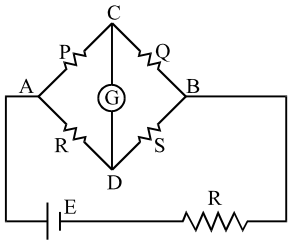
(a) emf = mE & **(b)** internal resistance = $\frac{mr}{n}$

Current $I = \frac{mE}{R+\frac{mr}{n}}$. For maximum current nR = mr or

$R = \frac{mr}{n}$ = internal resistance of battery .

$I_{max} = \frac{nE}{2r} = \frac{mE}{2R}$.

Wheat Stone Network :



When current through the galvanometer is zero (null point or balance point)

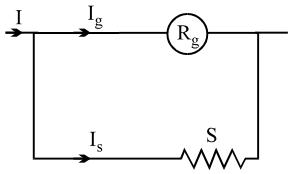
$\frac{P}{Q} = \frac{R}{S}$. When PS > QR; $V_C < V_D$ & PS <QR ; $V_C > V_D$ or PS = QR $\Rightarrow$ products of opposite arms are equal. Potential difference between C & D at null point is zero. The null point is not affected by resistance of G & E. It is not affected even if the positions of G & E are inter changed. $I_{CD} \alpha (QR - PS)$.

14. POTENTIOMETER :

A potentiometer is a linear conductor of uniform cross-section with a steady current set up in it. This maintains a uniform potential gradient along the length of the wire. Any potential difference which is less then the potential difference maintained across the potentiometer wire can be measured using this. The potentiometer equation is $\frac{E_1}{E_2} = \frac{I_1}{I_2}$.

15. AMMETER :

It is a modified form of suspended coil galvanometer it is used to measure current . A shunt (small resistance) is connected in parallel with galvanometer to convert into ammeter. $S = \frac{I_g R_g}{I - I_g}$;



An ideal ammeter has zero resistance. where

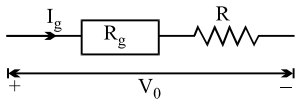
I_g = Maximum current that can flow through the galvanometer.

I = Maximum current that can be measured using the given ammeter.

16. VOLTMETER :

A high resistance is put in series with galvanometer . It is used to measure potential difference .

$I_g = \frac{V_o}{R_g + R}$.



$R \to \infty$, Ideal voltmeter .

17. RELATIVE POTENTIAL :

While solving an electric circuit it is convinient to chose a reference point and assigning its voltage as zero. Then all other potential are measured with respect to this point. This point is also called the common point.

18. ELECTRICAL POWER :

The energy liberated per second in a device is called its power. The electrical power P delivered by an electrical device is given by P = VI, where V = potential difference across device & I = current. If the current enters the higher potential point of the device then power is consumed by it (i.e. acts as load) . If the current enters the lower potential point then the device supplies power (i.e. acts as source) .

Power consumed by a resistor $P = I^2R = VI = \frac{V^2}{R}$.

19. HEATING EFFECT OF ELECTRIC CURRENT :

When a current is passed through a resistor energy is wested in over coming the resistances of the wire . This energy is converted into heat .

W = VIt Joule ; $= I^2Rt$ Joule ; $= \frac{V^2}{R}t$ Joule .

20. JOULES LAW OF ELECTRICAL HEATING :

The heat generated (in joules) when a current of I ampere flows through a resistance of R ohm for T second is given by :

$H = I^2RT$ Joules ; $= \frac{I^2RT}{4.2}$ Calories .

If current is variable passing through the conductor then we use for heat produced in resistance in time 0 to t is: $H = \int_0^t I^2R\,dt$

21. UNIT OF ELECTRICAL ENERGY CONSUMPTION :

1 unit of electrical energy = Kilowatt hour = 1 KWh = 3.6×10^6 Joules.

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1. Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the densities of charge carriers in the two wires are in the ratio 1 : 4, the drift velocity of electrons in the two wires will be in the ratio :

(A) 1 : 2 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

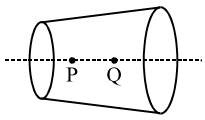
2. In a wire of cross-section radius r, free electrons travel with drift velocity v when a current I flows through the wire. What is the current in another wire of half the radius and of the same material when the drift velocity is 2v?

(A) 2I (B) I (C) $\frac{I}{2}$ (D) $\frac{I}{4}$

3. A current I flows through a uniform wire of diameter d when the mean electron drift velocity is V. The same current will flow through a wire of diameter d/2 made of the same material if the mean drift velocity of the electron is :

(A) $\frac{v}{4}$ (B) $\frac{v}{2}$ (C) 2v (D) 4v

4. A wire has a non-uniform cross-section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and q is v_P and v_Q.

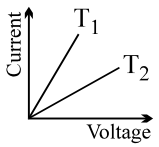


(A) $v_P = v_Q$ (B) $v_P < v_Q$ (C) $v_P > v_Q$ (D) Data insufficient

5. A uniform copper wire carries a current i amperes and has p carriers per metre^3. The length of the wire is ℓ metres and its cross-section area is s metre^2. If the charge on a carrier is q coulombs, the drift velocity in ms^{-1} is given by

(A) $\frac{i}{\ell sq}$ (B) $\frac{i}{psq}$ (C) $\frac{psq}{i}$ (D) $\frac{i}{ps\ell q}$

6. The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2. Which is correct



(A) $T_1 > T_2$ (B) $T_1 < T_2$ (C) $T_1 = T_2$ (D) none

7. A piece of copper and another of germanium are cooled from room temperature to 80K. The resistance of :

(A) each of them increases (B) each of them decreases

(C) copper increases and germanium decreases (D) copper decreases and germanium increases

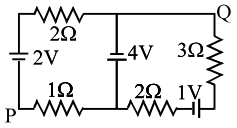
8. A brass disc and a carbon disc of same radius are assembled alternatively to make a cylindrical conductor. The resistance of the cylinder is independent of the temperature. The ratio of thickness of the brass disc to that of the carbon disc is [α is temperature coefficient of resistance & Neglect linear expansion]

(A) $\left|\frac{\alpha_C\rho_C}{\alpha_B\rho_B}\right|$ (B) $\left|\frac{\alpha_C\rho_B}{\alpha_B\rho_C}\right|$ (C) $\left|\frac{\alpha_B\rho_C}{\alpha_C\rho_B}\right|$ (D) $\left|\frac{\alpha_B\rho_B}{\alpha_C\rho_C}\right|$

9. A wire of resistance 10.0 ohm is stretched and increases its length by 20%. Its resistance then would be : (Neglect any change in volume or resistivity).

(A) 10.0 ohm (B) 12.0 ohm (C) 14.4 ohm (D) 10.2 ohm

10. In the circuit shown, what is the potential difference V_{PQ}?



(A) + 3V (B) + 2V (C) –2V (D) none

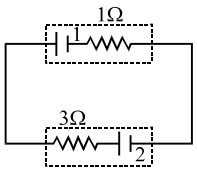
11. One end of a Nichrome wire of length 2L and cross-sectional area A is attatched to an end of another Nichrome wire of length L and cross-sectional area 2A. If the free end of the longer wire is at an electric potential of 8.0 volts, and the free end of the shorter wire is at an electric potential of 1.0 volt, the potential at the junction of the two wires is equal to

(A) 2.4 V (B) 3.2 V (C) 4.5 V (D) 5.6 V

12. A storage battery is connected to a charger for charging with a voltage of 12.5 Volts. The internal resistance of the storage battery is 1Ω. When the charging current is 0.5 A, the emf of the storage battery is :

(A) 13 Volts (B) 12.5 Volts (C) 12 Volts (D) 11.5 Volts

13. In the figure shown, battery 1 has emf = 6 V and internal resistance = 1 Ω. Battery 2 has emf = 2V and internal resistance = 3 Ω. The wires have negligible resistance. What is the potential difference across the terminals of battery 2 ?

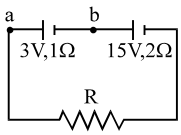


(A) 4 V (B) 1.5 V (C) 5 V (D) 0.5 V

14. The terminal voltage across a battery of emf E can be

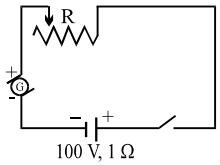
(A) 0 (B) > E (C) < E (D) all of above

15. Two batteries one of the emf 3V, internal resistance 1 ohm and the other of emf 15 V, internal resistance 2 ohm are connected in series with a resistance R as shown. If the potential difference between a and b is zero the resistance of R in ohm is



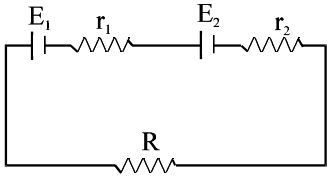
(A) 5 (B) 7 (C) 3 (D) 1

16. The battery in the diagram is to be charged by the generator G. The generator has a terminal voltage of 120 volts when the charging current is 10 amperes. The battery has an emf of 100 volts and an internal resistance of 1 ohm. In order to charge the battery at 10 amperes charging current, the resistance R should be set at



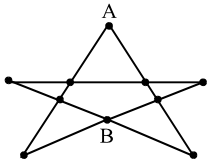
(A) 0.1 Ω (B) 0.5 Ω (C) 1.0 Ω (D) 5.0 Ω

17. Under what condition current passing through the resistance R can be increased by short circuiting the battery of emf E_2. The internal resistances of the two batteries are r_1 and r_2 respectively.



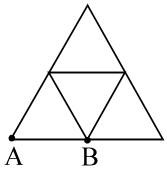
(A) $E_2r_1 > E_1 (R + r_2)$ (B) $E_1r_2 > E_2 (R + r_1)$
(C) $E_2r_2 > E_1 (R + r_2)$ (D) $E_1r_1 > E_2 (R + r_1)$

18. The resistance of all the wires between any two adjacent dots is R. Then equivalent resistance between A and B as shown in figure is :



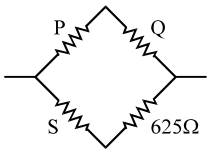
(A) $\frac{7}{3}R$ (B) $\frac{7}{6}R$ (C) $\frac{14}{8}R$ (D) None of these

19. In the diagram resistance between any two junctions is R. Equivalent resistance across terminals A and B is



(A) $\frac{11R}{7}$ (B) $\frac{18R}{11}$ (C) $\frac{7R}{11}$ (D) $\frac{11R}{18}$

20. A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51Ω to secure balance. The unknown resistance in the fourth arm is



(A) 625 Ω (B) 650 Ω (C) 676 Ω (D) 600 Ω

21. In a balanced wheat stone bridge, current in the galvanometer is zero. It remains zero when :

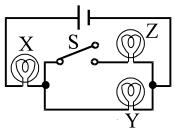
[1] battery emf is increased [2] all resistances are increased by 10 ohms

[3] all resistances are made five times [4] the battery and the galvanometer are interchanged

(A) only [1] is correct (B) [1], [2] and [3] are correct

(C) [1], [3] and [4] are correct (D) [1] and [3] are correct

22. If X, Y and Z in figure are identical lamps, which of the following changes to the brightnesses of the lamps occur when switch S is closed?



(A) X stays the same, Y decreases (B) X increases, Y decreases

(C) X increases, Y stays the same (D) X decreases, Y increases

23. Power generated across a uniform wire connected across a supply is H. If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

(A) $\frac{H}{n^2}$ (B) n^2H (C) nH (D) $\frac{H}{n}$

24. A constant voltage is applied between the two ends of a uniform metallic wire. Some heat is developed in it. The heat developed is doubled if

(A) both the length and the radius of the wire are halved.

(B) both the length and the radius of the wire are doubled

(C) the radius of the wire is doubled

(D) the length of the wire is doubled

25. Two bulbs rated (25 W – 220V) and (100W – 220V) are connected in series to a 440 V line. Which one is likely to fuse?

(A) 25 W bulb (B) 100 W bulb (C) both bulbs (D) none

26. Rate of dissipation of Joule's heat in resistance per unit volume is (symbols have usual meaning)

(A) σE (B) σJ (C) J E (D) None

27. If the length of the filament of a heater is reduced by 10%, the power of the heater will

(A) increase by about 9% (B) increase by about 11%

(C) increase by about 19% (D) decrease by about 10%

28. A heater A gives out 300 W of heat when connected to a 200 V d.c. supply. A second heater B gives out 600 W when connected to a 200 v d.c. supply. If a series combination of the two heaters is connected to a 200 V d.c. supply the heat output will be

(A) 100 W (B) 450 W (C) 300 W (D) 200 W

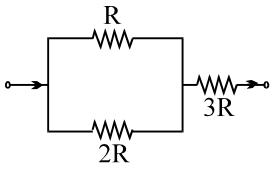
29. Two bulbs one of 200 volts, 60 watts & the other of 200 volts, 100 watts are connected in series to a 200 volt supply. The power consumed will be

(A) 37.5 watt (B) 160 watt (C) 62.5 watt (D) 110 watt

30. Three 60 W light bulbs are mistakenly wired in series and connected to a 120 V power supply. Assume the light bulbs are rated for single connection to 120 V. With the mistaken connection, the power dissipated by each bulb is :

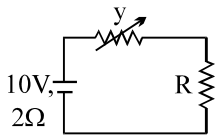
(A) 6.7 W (B) 13.3 W (C) 20 W (D) 40 W

31. The ratio of powers dissipatted respectively in R and 3R, as shown is :



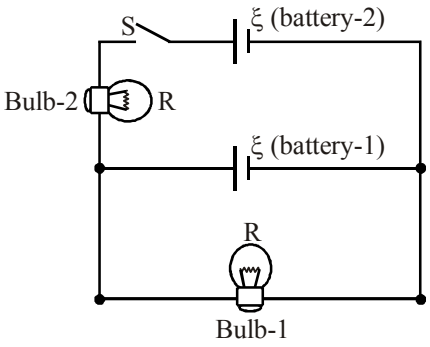
(A) 9 (B) $\frac{27}{4}$ (C) $\frac{4}{9}$ (D) $\frac{4}{27}$

32. In the figure shown the power generated in y is maximum when y = 5Ω. Then R is



(A) 2 Ω (B) 6 Ω (C) 5 Ω (D) 3 Ω

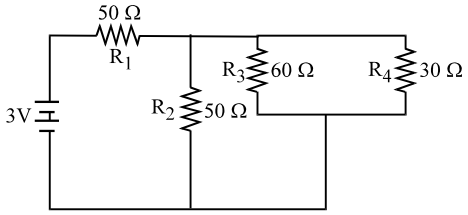
33. For the situation shown, the switch labeled S in the circuit is originally open. Assume that the batteries have no internal resistance, identical emf, and the light bulbs have equal resistance. Which statement is correct after closing the switch?



(A) Bulb-1 brightens (B) Battery-2 provides more power than Battery-1

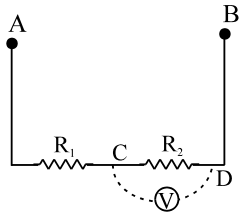
(C) Bulb-2 is brighter than Bulb-1 (D) The current from Battery-1 is unchanged

34. In the circuit shown, the resistances are given in ohms and the battery is assumed ideal with emf equal to 3.0 volts. The resistor that dissipates the most power is



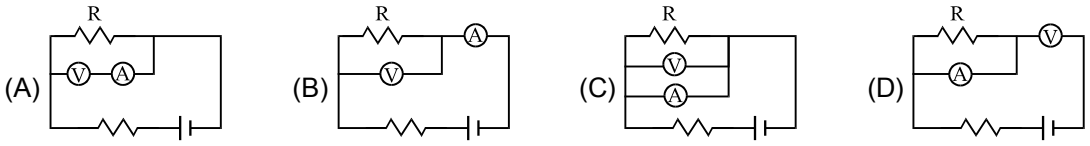
(A) R_1 (B) R_2 (C) R_3 (D) R_4

35. Resistances R_1 and R_2 each 60Ω are connected in series as shown in figure. The Potential difference between A and B is kept 120 volt. Then what will be the reading of voltmeter connected between the point C & D if resistance of voltmeter is 120Ω.

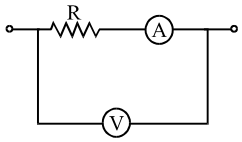


(A) 48 V (B) 24 V (C) 40V (D) None

36. Which of the following wiring diagrams could be used to experimentally determine R using ohm's law? Assume an ideal voltmeter and an ideal ammeter.

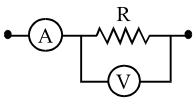


37. In the circuit shown the resistance of voltmeter is 10,000 ohm and that of ammeter is 20 ohm. The ammeter reading is 0.10 Amp and voltmeter reading is 12 volt. Then R is equal to



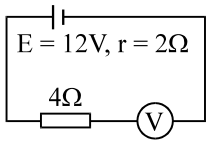
(A) 122 Ω (B) 140 Ω (C) 116 Ω (D)100 Ω

38. In the circuit shown the readings of ammeter and voltmeter are 4A and 20V respectively. The meters are non ideal, then R is :



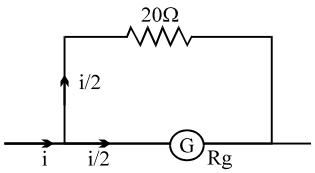
(A) 5Ω (B) less than 5Ω (C) greater than 5Ω (D) between 4Ω & 5Ω

39. By error, a student places moving-coil voltmeter V (nearly ideal) in series with the resistance in a circuit in order to read the current, as shown. The voltmeter reading will be



(A) 0 (B) 4V (C) 6V (D) 12V

40. In a galvanometer, the deflection becomes one half when the galvanometer is shunted by a 20Ω resistor. The galvanometer resistance is



(A) 5Ω (B) 10Ω (C) 40Ω (D) 20Ω

41. A galvanometer has a resistance of 20Ω and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

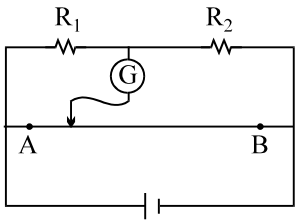
(A) 0.01Ω resistor connected across it (B) 0.02Ω resistor connected across it

(C) 200Ω resistor connected in series with it (D) 2000 Ω resistor connected in series with it

42. A galvanometer coil has a resistance 90 Ω and full scale deflection current 10 mA. A 910Ω resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1V, the number of divisions on its scale is

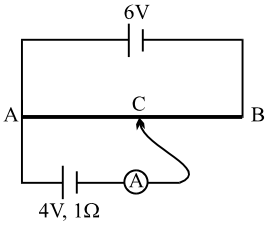
(A) 90 (B) 91 (C) 100 (D) none

43. In the figure shown for gives values of R_1 and R_2 the balance point for Jockey is at 40 cm from A. When R_2 is shunted by a resistance of 10 Ω, balance shifts to 50 cm. R_1 and R_2 are (AB = 1 m) :



(A) $\frac{10}{3}$ Ω, 5 Ω (B) 20 Ω, 30 Ω (C) 10 Ω, 15 Ω (D) 5 Ω, $\frac{15}{2}$ Ω

44. A 6 V battery of negligible internal resistance is connected across a uniform wire of length 1 m. The positive terminal of another battery of emf 4V and internal resistance 1 Ω is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. The AC is equal to



(A) $\frac{2}{3}$m (B) $\frac{1}{3}$m (C) $\frac{3}{5}$m (D) $\frac{1}{2}$m

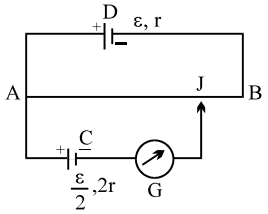
45. A potentiometer wire has length 10 m and resistance 10Ω. It is connected to a battery of EMF 11 volt and internal resistance 1 Ω, then the potential gradient in the wire is

(A) 10 V/m (B) 1 V/m (C) 0.1 V/m (D) none

46. The length of a potentiometer wire is *l*. A cell of emf E is balanced at a length *l*/3 from the positive end of the wire. If the length of the wire is increased by *l*/2. At what distance will the same cell give a balance point.

(A) $\frac{2l}{3}$ (B) $\frac{l}{2}$ (C) $\frac{l}{6}$ (D) $\frac{4l}{3}$

47. In the figure, the potentiometer wire AB of length L and resistance 9r is joined to the cell D of emf ε and internal resistance r. The cell C's emf is ε/2 and its internal resistance is 2r. The galvanometer G will show no deflection when the length AJ is



(A) $\frac{4L}{9}$ (B) $\frac{5L}{9}$ (C) $\frac{7L}{18}$ (D) $\frac{11L}{18}$

48. In the given potentiometer circuit length of the wire AB is 3 m and resistance is R = 4.5 Ω. The length AC for no deflection in galvanometer is

(A) 2 m (B) 1.8 m

(C) dependent on r_1 (D) none of these

49. An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions → their charges being –e and +2e. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v (–ve ion) and v/4 (+ve ion). Both ions have the same number per unit volume = n. The current flowing through the pipe is

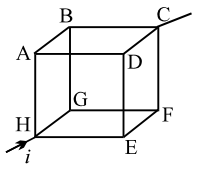
(A) nev A/2 (B) nev A/4 (C) 5nev A/2 (D) 3nev A/2

50. The total momentum of electrons in a straight wire of length 1000 m carrying a current of 70 A is closest to.

(A) 40×10^{-8} N-sec (B) 30×10^{-8} N-sec

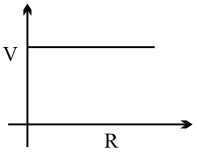
(C) 50×10^{-8} N-sec (D) 70×10^{-8} N-sec

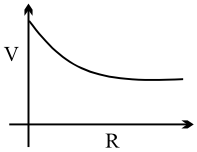
51. In the box shown current i enters at H and leaves at C. If $i_{AB} = \frac{i}{6}$, $i_{DC} = \frac{2i}{3}$, $i_{HA} = \frac{i}{2}$, $i_{GF} = \frac{i}{6}$, $i_{HE} = \frac{i}{6}$, choose the branch in which current is zero

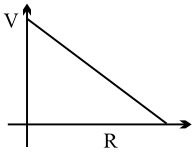


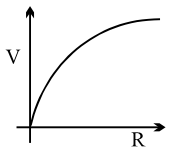
(A) BG (B) FC (C) ED (D) none

52. A cell of emf E has an internal resistance r and is connected to rheostat. When resistance R of rheostat is changed correct graph of potential difference across it is

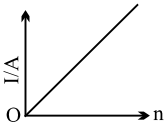
(A) 

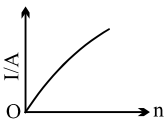
(B) 

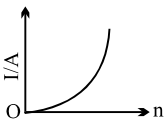
(C) 

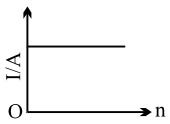
(D) 

53. A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured. Which one of the graph below shows the relationship between I and n?

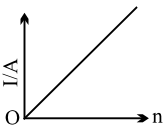
(A) 

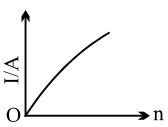
(B) 

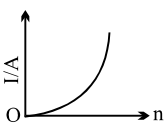
(C) 

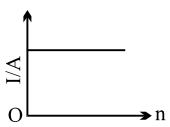
(D) 

54. In **previous problem**, if the cell had been connected in parallel (instead of in series) which of the above graphs would have shown the relationship between total current I and n?

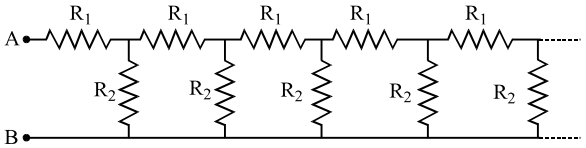
(A) 

(B) 

(C) 

(D) 

55. Consider an infinte ladder network shown in figure. A voltage V is applied between the points A and B. This applied value of voltage is halved after each section.



(A) $\frac{R_1}{R_2} = 1$ (B) $\frac{R_1}{R_2} = \frac{1}{2}$ (C) $\frac{R_1}{R_2} = 2$ (D) $\frac{R_1}{R_2} = 3$

56. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current, the temperature of the wire is raised by ΔT in time t. N number of similar cells is now connected in series with a wire of the same material and cross section but of length 2L. The temperature of the wire is raised by the same amount ΔT in the same time t. The value of N is :

(A) 4 (B) 6 (C) 8 (D) 9

57. When electric bulbs of same power, but different marked voltage are connected in series across the power line, their brightness will be :

(A) proportional to their marked voltage

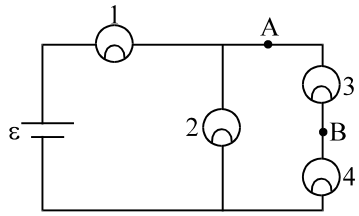
(B) inversely proportional to their marked voltage

(C) proportional to the square of their marked voltage

(D) inversely proportional to the square of their marked voltage

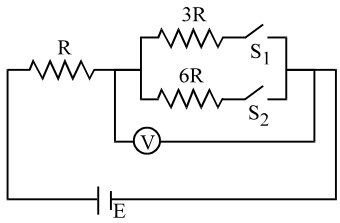
(E) the same for all of them

58. For the circuit shown, a shorting wire of negligible resistance is added to the circuit between points A and B. When this shorting wire is added, bulb-3 goes out. Which bulbs (all identical) in the circuit brighten?



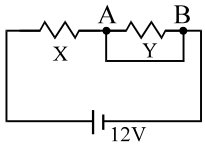
(A) only bulb 2 (B) only bulb 4 (C) only bulbs 1 and 4 (D) only bulbs 2 and 4

59. In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed. The reading of voltmeter is V_3 when both S_1 and S_2 are closed then



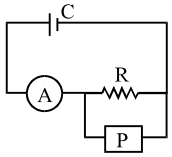
(A) $V_2 > V_1 > V_3$ (B) $V_3 > V_2 > V_1$ (C) $V_3 > V_1 > V_2$ (D) $V_1 > V_2 > V_3$

60. When an ammeter of negligible internal resistance is inserted in series with circuit it reads 1A. When the voltmeter of very large resistance is connected across X it reads 1V. When the point A and B are shorted by a conducting wire, the voltmeter measures 10 V across the battery. The internal resistance of the battery is equal to



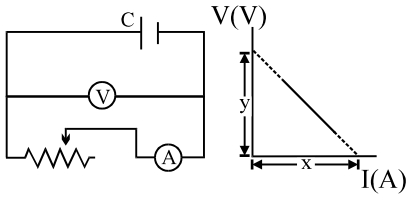
(A) zero (B) 0.5 Ω (C) 0.2 Ω (D) 0.1 Ω

61. An ammeter A of finite resistance, and a resistor R are joined in series to an ideal cell C. A potentiometer P is joined in parallel to R. The ammeter reading is I_0 and the potentiometer reading is V_0. P is now replaced by a voltmeter of finite resistance. The ammeter reading now is I and the voltmeter reading is V.



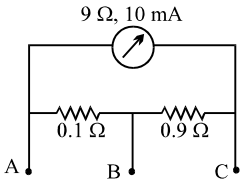
(A) $I > I_0$, $V < V_0$ (B) $I > I_0$, $V = V_0$ (C) $I = I_0$, $V < V_0$ (D) $I < I_0$, $V = V_0$

62. The diagram besides shows a circuit used in an experiment to determine the emf and internal resistance of the cell C. A graph was plotted of the potential difference V between the terminals of the cell against the current I, which was varied by adjusting the rheostat. The graph is shown on the right ; x and y are the intercepts of the graph with the axes as shown. What is the internal resistance of the cell?



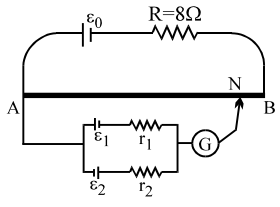
(A) x (B) y (C) $\frac{x}{y}$ (D) $\frac{y}{x}$

63. A milliammeter of range 10 mA and resistance 9 Ω is joined in a circuit as shown. The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B (C is left isolated). The value of I is



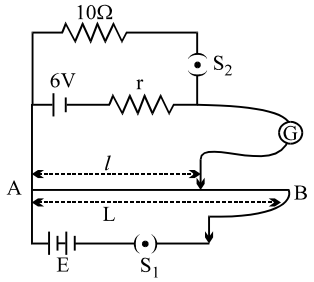
(A) 100 mA (B) 900 mA (C) 1 A (D) 1.1 A

64. A battery of emf E_0 = 12 V is connected across a 4m long uniform wire having resistance 4Ω/m. The cells of small emfs ε_1 = 2V and ε_2 = 4V having internal resistance 2Ω and 6Ω respectively, are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to



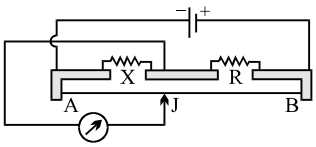
(A) $\frac{1}{6}$ m (B) $\frac{1}{3}$ m (C) 25 cm (D) 50 cm

65. In the arrangement shown in figure when the switch S_2 is open, the galvanometer shows no deflection for l = L/2. When the switch S_2 is closed, the galvanometer shows no deflection for l = 5L / 12. The internal resistance (r) of 6 V cell, and the emf E of the other battery are respectively



(A) 3Ω, 8V (B) 2Ω, 12V (C) 2Ω, 24V (D) 3Ω, 12V

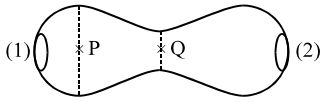
66. The figure shows a metre-bridge circuit, with AB = 100 cm, X = 12Ω and R = 18Ω, and the jockey J in the position of balance. If R is now made 8Ω, through what distance will J have to be moved to obtain balance?



(A) 10 cm (B) 20 cm (C) 30 cm (D) 40 cm

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

67. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



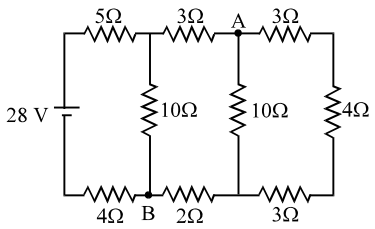
(A) the current at the cross-section P equals the current at the cross-section Q
(B) the electric field intensity at P is less than that at Q.
(C) the rate of heat generated per unit time at Q is greater than that at P
(D) the number of electrons crossing per unit area of cross-section at P is less than that at Q.

68. Which of the following quantities do not change when an ohmic resistor connected to a battery is heated due to the current?
(A) drift speed (B) resistivity (C) resistance (D) number of free electrons

69. A current passes through an ohmic conductor of nonuniform cross section. Which of the following quantities are independent of the cross-section?
(A) the charge crossing in a given time interval. (B) drift speed
(C) current density (D) free-electron density

70. A simple circuit contains an ideal battery and a resistance R. If a second resistor is placed in parallel with the first,
(A) the potential across R will decrease
(B) the current through R will decreased
(C) the current delivered by the battery will increase
(D) the power dissipated by R will increased.

71. Consider the circuit shown in the figure



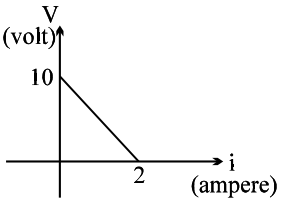
(A) the current in the 5 Ω resistor is 2 A (B) the current in the 5 Ω resistor is 1 A
(C) the potential difference $V_A - V_B$ is 10 V (D) the potential difference $V_A - V_B$ is 5 V

72. A battery is of emf E is being charged from a charger such that positive terminal of the battery is connected to terminal A of charger and negative terminal of the battery is connected to terminal B of charger. The internal resistance of the battery is r.
(A) Potential difference across points A and B must be more than E.
(B) A must be at higher potential than B
(C) In battery, current flows from positive terminal to the negative terminal
(D) No current flows through battery

73. The equivalent resistance of a group of resistances is R. If another resistance is connected in parallel to the group, its new equivalent becomes R_1 and if it is connected in series to the group, its new equivalent becomes R_2 we have :
(A) $R_1 > R$ (B) $R_1 < R$ (C) $R_2 > R$ (D) $R_2 < R$

74. Two identical fuses are rated at 10A. If they are joined
(A) in parallel, the combination acts as a fuse of rating 20A
(B) in parallel, the combination acts as a fuse of rating 5A
(C) in series, the combination acts as a fuse of rating 10A.
(D) in series, the combination acts as a fuse of rating 20A.

75. A battery of emf E and internal resistance r is connected across a resistance R. Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative(s).



(A) internal resistance of battery is 5Ω
(B) emf of the battery is 20V
(C) maximum current which can be taken from the battery is 4A
(D) V- i graph can never be a straight line as shown in figure.

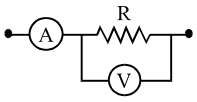
76. A galvanometer may be converted into ammeter or voltmeter. In which of the following cases the resistance of the device will be the largest? (Asssume maximum range of galvanometer = 1 mA)
(A) an ammeter of range 10A (B) a voltmeter of range 5 V
(C) an ammeter of range 5 A (D) a voltmeter of range 10 V.

77. Mark out the correct options.

(A) An ammeter should have small resistance. (B) An ammeter should have large resistance.

(C) A voltmeter should have small resistance. (D) A voltmeter should have large resistance.

78. In the circuit shown the readings of ammeter and voltmeter are 4A and 20V respectively. The meters are non ideal, then R is :



(A) 5Ω (B) less than 5Ω

(C) greater than 5Ω (D) between 4Ω & 5Ω

79. A micrometer has a resistance of 100Ω and a full scale range of 50 μA. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).

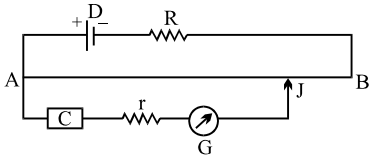
(A) 50 V range with 10 kΩ resistance in series. (B) 10 V range with 200 kΩ resistance in series.

(C) 5 mA range with 1 Ω resistance in parallel. (D) 10 mA range with 1 kΩ resistance in parallel.

80. In a potentiometer wire experiment the emf of a battery in the primary circuit is 20V and its internal resistance is 5Ω. There is a resistance box in series with the battery and the potentiometer wire, whose resistance can be varied from 120Ω to 170Ω. Resistance of the potentiometer wire is 75Ω. The following potential differences can be measured using this potentiometer.

(A) 5V (B) 6V (C) 7V (D) 8V

81. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0. C is a cell of internal resistance r. The galvanometer G does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this?



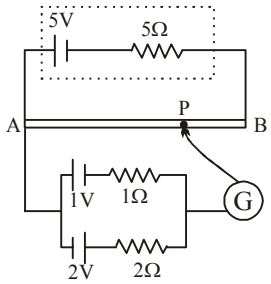
(A) $r > R_0$

(B) $R >> R_0$

(C) emf of C > emf of D

(D) The negative terminal of C is connected to A.

82. A battery of emf ε_0 = 5V & internal resistance 5Ω is connected across a long uniform wire AB of length 1 m & resistance per unit length 5 Ωm^{-1}. Two cells of ε_1 = 1V & ε_2 = 2V are connected as shown in the fig.



(A) The null point is at A

(B) If the Jockey is touched to point B the current in the galvanometer will be going towards B

(C) When Jockey is connected to point A no current is flowing through 1 V battery

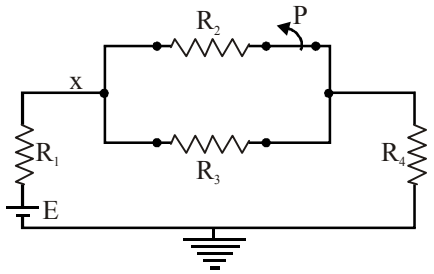
(D) The null point is at distance of 8/15 m from A

83. The current in a wire is doubled keeping temperature constant. Which of the following statements is/are wrong?

(A) The current density is doubled (B) The conduction electron density is doubled

(C) The mean time between collisions is constant (D) The electron drift speed is doubled

84. If the switch at point P is opened (shown in the figure) choose the correct option



(A) the current in R_1 would not change
(B) the potential difference between point X and the ground would increase
(C) the current provided by the battery would increase
(D) the emf produced by the battery (assumed to have no internal resistance) would change

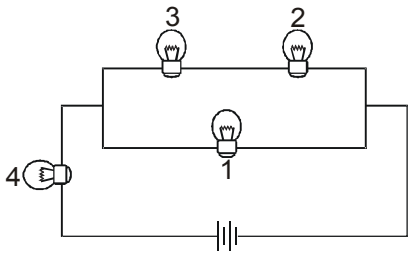
85. **Statement-1 :** The resistivity of a semiconductor decreases with increases of temperature.
Statement-2 : In a conducting solid, the rate of collision between free electrons and ions increases with increase of temperature.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

86. **Statement-1 :** When two conducting wires of different resistivity having same cross section area are joined in series, the electric field in them would be equal when they carry current.
Statement-2 : When wires are in series they carry equal current.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

87. **Statement-1 :** Potential difference across the terminals of a battery is always less than its emf.
Statement-2 : A battery always has some internal resistance.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

Paragraph for question nos. 88 to 90

All bulbs consume same power. The resistance of bulb 1 is 36Ω.



88. What is the resistance of bulb 3 :
(A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω

89. What is the resistance of bulb 4 :
(A) 4Ω (B) 9Ω (C) 12Ω (D) 18Ω

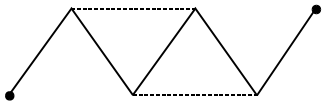
90. What is the voltage output of the battery if the power of each bulb is 4W :
(A) 12V (B) 16V (C) 24V (D) None of these

EXERCISE-II

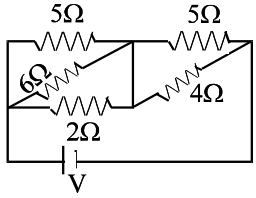
1. **(a)** The current density across a cylindrical conductor of radius R varies according to the equation $J = J_0\left(1-\frac{r}{R}\right)$, where r is the distance from the axis. Thus the current density is a maximum J_o at the axis r = 0 and decreases linearly to zero at the surface r = R. Calculate the current in terms of J_o and the conductor's cross sectional area is $A = \pi R^2$.

(b) Suppose that instead the current density is a maximum J_o at the surface and decreases linearly to zero at the axis so that $J = J_0\frac{r}{R}$. Calculate the current.

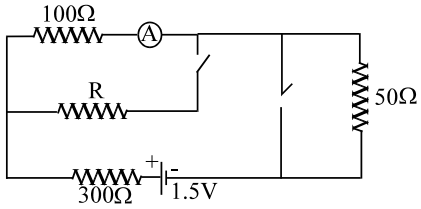
2. By what factor will the resistance of a circuit consisting of five identical conductors change if two similar conductors are added as shown by the dashed line in figure.



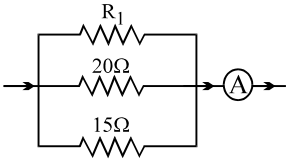
3. Find the resistor in which maximum heat will be produced.



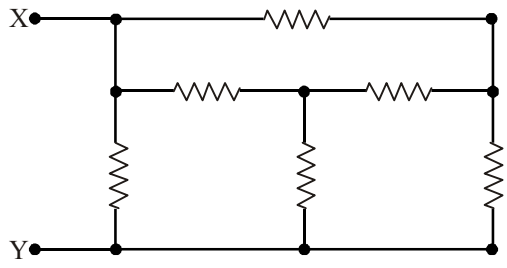
4. In the circuit shown in figure the reading of ammeter is the same with both switches open as with both closed. Then find the resistance R. (ammeter is ideal)



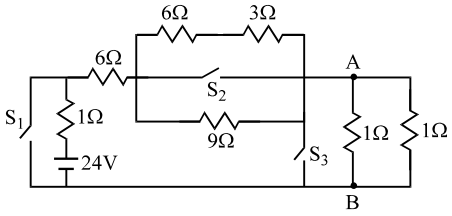
5. In the given circuit the current flowing through the resisitance 20 ohms is 0.3 ampere while the ammetre reads 0.8 ampere. What is the value of R_1?



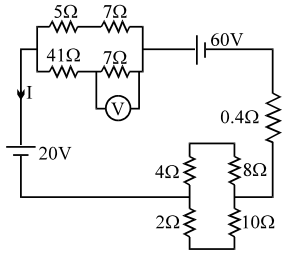
6. Six resistors each of 10 Ω are connected as shown. Find the equivalent resistance between points X and Y.



7. If the switches S_1, S_2 and S_3 in the figure are arranged such that current through the battery is minimum, find the voltage across points A and B.

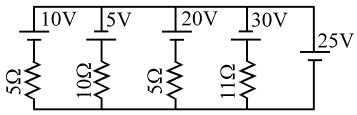


8. Find the current I & voltage V in the circuit shown.

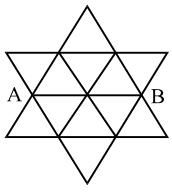


9. If a cell of constant E.M.F. produces the same amount of the heat during the same time in two independent resistors R_1 and R_2, when they are separately connected across the terminals of the cell, one after the another, find the internal resistance of the cell.

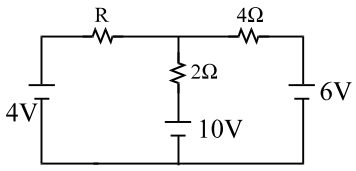
10. Find the current through 25V cell & power supplied by 20V cell in the figure shown.



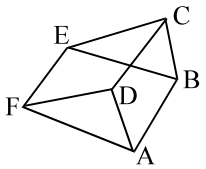
11. Find the equivalent resistance of the circuit between points A and B shown in figure is :
(each branch is of resistance = 1Ω)



12. For what value of R in circuit, current through 4Ω resistance is zero.

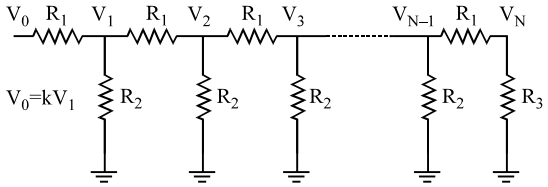


13. In the circuit shown in figure, all wires have equal resistance r. Find the equivalent resistance between A and B.



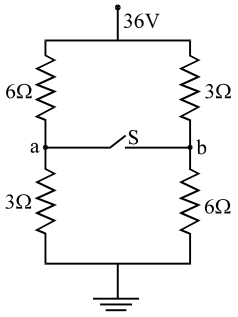
14. The current I through a rod of a certain metallic oxide is given by $I = 0.2\ V^{5/2}$, where V is the potential difference across it. The rod is connected in series with a resistance to a 6V battery of negligible internal resistance. What value should the series resistance have so that :
(i) the current in the circuit is 0.44;
(ii) the power dissipated in the rod is twice that dissipated in the resistance.

15. A network of resistance is constructed with R_1 & R_2 as shown in the figure. The potential at the points 1, 2, 3,.., N are V_1, V_2, V_3 , .., V_n respectively each having a potential k time smaller than previous one. Find :
(i) $\frac{R_1}{R_2}$ and $\frac{R_2}{R_3}$ in terms of k
(ii) current that passes through the resistance R_2 nearest to the V_0 in terms V_0, k & R_3.



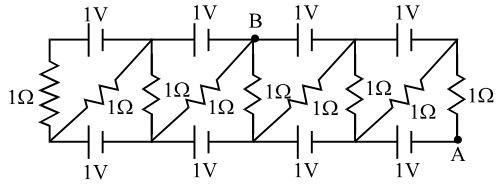
16. A person decides to use his bath tub water to generate electric power to run a 40 watt bulb. The bath tube is located at a height of 10 m from the ground & it holds 200 litres of water. If we install a water driven wheel generator on the ground, at what rate should the water drain from the bath tube to light bulb? How long can we keep the bulb on, if the bath tub was full initially. The efficiency of generator is 90%.($g = 10\ m/s^{-2}$)

17. In the circuit shown in figure, calculate the following :
(i) Potential difference between points a and b when switch S is open.
(ii) Current through S in the circuit when S is closed.

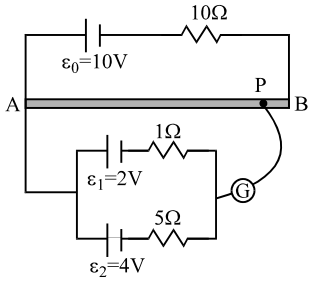


18. An ideal cell having a steady emf of 2 volt is connected across the potentiometer wire of length 10 m. The potentiometer wire is of magnesium and having resistance of 11.5 Ω/m. An another cell gives a null point at 6.9 m. If a resistance of 5Ω is put in series with potentiometer wire, find the new position of the null point.

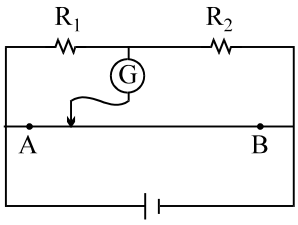
19. Find the potential difference $V_A - V_B$ for the circuit shown in the figure.



20. A battery of emf ε_0 = 10 V is connected across a 1 m long uniform wire having resistance 10 Ω/m. Two cells of emf ε_1 = 2V and ε_2 = 4V having internal resistances 1Ω and 5Ω respectively are connected as shown in the figure. If a galvanometer shows no deflection at the point P, find the distance of point P from the point a.



21. In the figure shown for which values of R_1 and R_2 the balance point for Jockey is at 40 cm from A. When R_2 is shunted by a resistance of 10 Ω, balance shifts to 50 cm. Find R_1 and R_2. (AB = 1 m) :



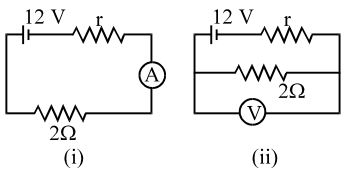
22. An accumulator of emf 2 Volt and negligible internal resistance is connected across a uniform wire of length 10m and resistance 30Ω. The appropriate terminals of a cell of emf 1.5 Volt and internal resistance 1Ω is connected to one end of the wire, and the other terminal of the cell is connected through a sensitive galvanometer to a slider on the wire. What length of the wire will be required to produce zero deflection of the galvanometer? How will the balancing change (a) when a coil of resistance 5Ω is placed in series with the accumulator, (b) the cell of 1.5 volt is shunted with 5Ω resistor?

23. A galvanometer (coil resistance 99 Ω) is converted into a ammeter using a shunt of 1Ω and connected as shown in the figure (i). The ammeter reads 3A. The same galvanometer is converted into a voltmeter by connecting a resistance of 101 Ω in series. This voltmeter is connected as shown in figure(ii). Its reading is found to be 4/5 of the full scale reading. Find

(a) internal resistance r of the cell

(b) range of the ammeter and voltmeter

(c) full scale deflection current of the galvanometer



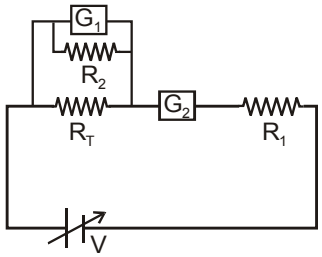
EXERCISE-III

1. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100 W, 60 W and 40 W bulbs have filament resistances R_{100}, R_{60} and R_{40}, respectively, the relation between these resistances is **[JEE 2010]**

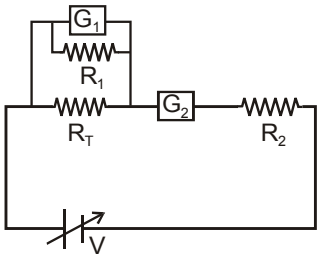
(A) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$ (B) $R_{100} = R_{40} + R_{60}$ (C) $R_{100} > R_{60} > R_{40}$ (D) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

2. To Verify Ohm's law, a student is provided with a test resistor R_T, a high resistance R_1, a small resistance R_2, two identical galvanometers G_1 and G_2, and a variable voltage source V. The correct circuit to carry out the experiment is **[JEE 2010]**

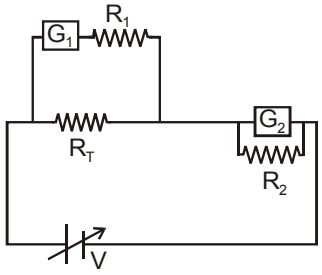
(A)



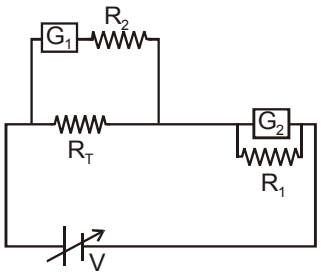
(B)



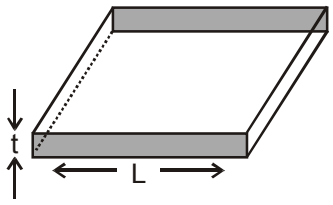
(C)



(D)



3. Consider a thin square sheet of side L and thickness t, made of material of resistivity ρ. The resistance between two opposite faces, shown by the shaded areas in the figure is **[JEE 2010]**

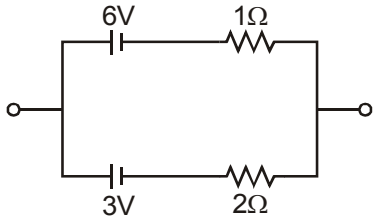


(A) directly proportional to L (B) directly proportional to t

(C) indenpendent of L (D) independent of t

4. When two identical batteries of internal resistance 1Ω each are connected in series across a resistor R, the rate of heat produced in R is J_1. When the same batteries are connected in parallel across R, the rate is J_2. If $J_1 = 2.25\ J_2$ then the value of R in Ω is **[JEE 2010]**

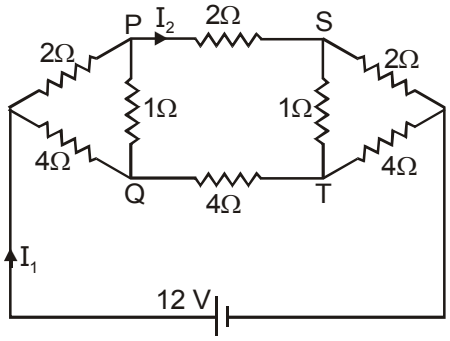
5. Two batteries of different emfs and different internal resistances are connected as shown. The voltage across AB is volts is **[JEE 2011]**



6. Two large vertical and parallel metal plates having a separation of 1 cm are connected to a DC voltage source of potential difference X. A proton is released at rest midway between the two plates. It is found to move at 45° to the vertical JUST after release. Then X is nearly **[JEE 2012]**

(A) 1×10^{-5} V (B) 1×10^{-7} V (C) 1×10^{-9} V (D) 1×10^{-10} V

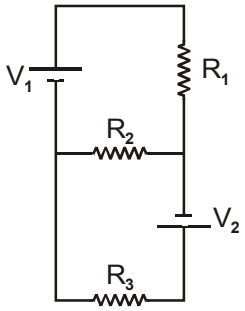
7. For the resistance network shown in the figure, choose the correct option(s). **[JEE 2012]**



(A) The current through PQ is zero (B) $I_1 = 3A$

(C) The potential at S is less than that at Q (D) $I_2 = 2A$

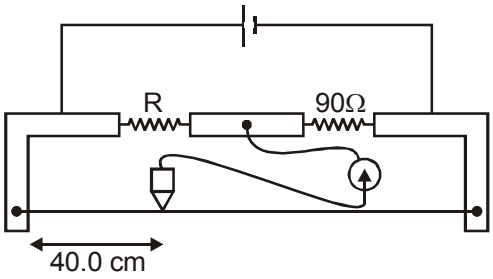
8. Two ideal batteries of emf V_1 and V_2 and three resistances R_1, R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if **[IIT-JEE Advance 2014]**



(A) $V_1 = V_2$ and $R_1 = R_2 = R_3$ (B) $V_1 = V_2$ and $R_1 = 2R_2 = R_3$

(C) $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$ (D) $2V_1 = V_2$ and $2R_1 = R_2 = R_3$

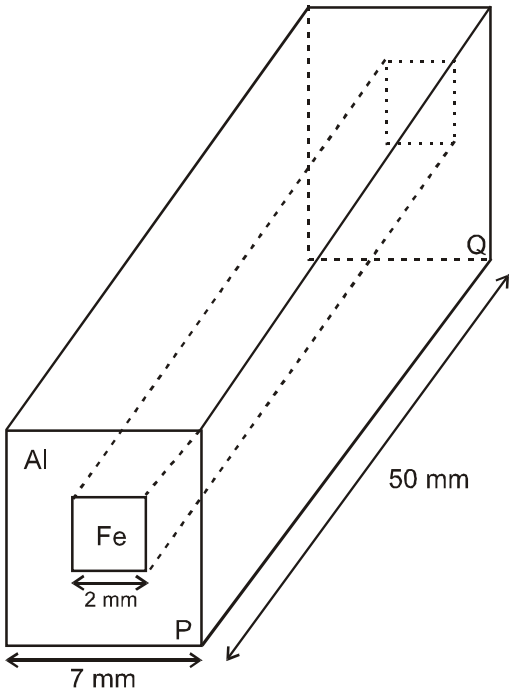
9. During an experiment with a metre bridge, the galvanometer shows a null point when the jockey is pressed at 40.0 cm using a standard resistance of 90 Ω, as shown in the figure. The least count of the scale used in the metre bridge is 1 mm. The unknown resistance is **[IIT-JEE Advance 2014]**



(A) 60 ± 0.15Ω (B) 135 ± 0.56Ω (C) 60 ± 0.25Ω (D) 135 ± 0.23Ω

10. A galvanometer gives full scale deflection with 0.006 A current. By connecting it to a 4990 Ω resistance, it can be converted into a voltmeter of range 0 – 30 V. If connected to a $\frac{2n}{249}\Omega$ resistance, it becomes an ammeter of range 0 – 1.5 A. The value of n is **[IIT-JEE Advance 2014]**

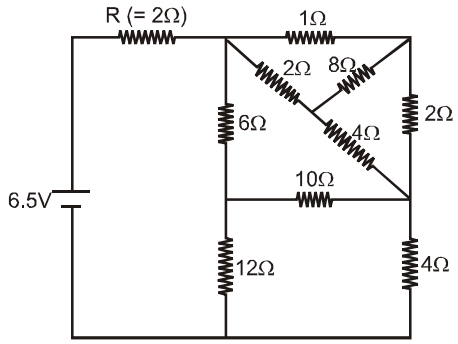
11. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are 2.7×10^{-8} Ωm and 1.0×10^{-7} Ωm, respectively. The electrical resistance between the two faces P and Q of the composite bar is **[IIT-JEE Advance 2015]**



(A) $\frac{2475}{64}\mu\Omega$ (B) $\frac{1875}{64}\mu\Omega$ (C) $\frac{1875}{49}\mu\Omega$ (D) $\frac{2475}{132}\mu\Omega$

12. In the following circuit, the current through the resistor R (= 2Ω) is I Amperes. The value of I is

[IIT-JEE Advance 2015]



13. Consider two identical galvanometers and two identical resistors with resistance R. If the internal resistance of the galvanometers $R_C < R/2$, which of the following statement(s) about any one of the galvanometers is(are) true ? **[IIT-JEE Advance 2016]**

(A) The maximum voltage range is obtained when all the components are connected in series.

(B) The maximum voltage range is obtained when the two resistors and one galvanomete are connected in series, and the second galvaanometer is connected in parallel to the first galvanometer.

(C) The maximum current range is obtained when all the components are connected in parallel.

(D) The maximum current range is obtained when the two galvanometers are connected in series and the combination is connected in parallel which both the resistors.

14. Two identical moving coil galvanometers have 10Ω resistance and full scale deflection at 2μA current. One of them is converted into a voltmeter of 100 mV full scale reading and the other into an Ammeter of 1mA full scale current using appropriate resistors. These are then used to measrue the voltage and current in the Ohm's law experiment with R = 1000 Ω resistor by using an ideal cell. Which of the following statement(s) is/are correct ? **[IIT-JEE Advance 2019]**

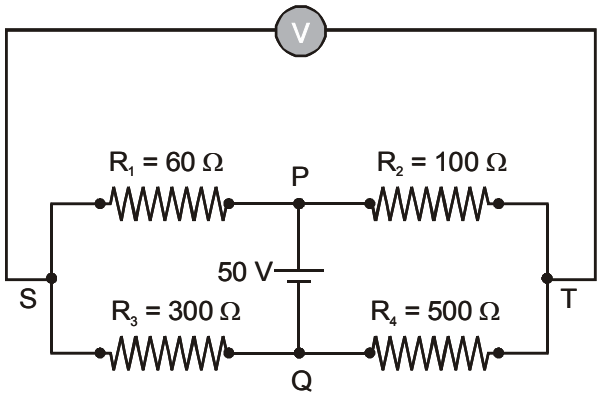
(A) The resistance of the Voltmeter will be 100 kΩ

(B) If the ideal cell is replaced by a cell having internal resistance of 5Ω then the measured value of R will be more than 1000 Ω

(C) The resistance of the Ammeter will be 0.02 Ω (round off to 2nd decimal place)

(D) The measured value of R will be 978 Ω < R < 982 Ω

15. In the balanced condition, the values of the resistances of the four arms of a Wheatstone bridge are shown in the figure below. The resistance R_3 has temperature coefficient 0.0004 $°C^{-1}$. If the temperature of R_3 is increased by 100 °C, the voltage developed between S and T will be _____ volt. **[IIT-JEE Advance 2020]**

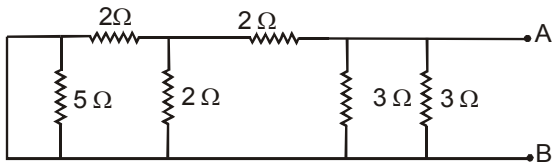


16. An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W.

[JEE-Mains Online-2021]

(A) 20 Ω (B) 5 Ω (C) 30 Ω (D) 10 Ω

17. The equivalent resistance of the given circuit between the terminals A and B is : **[JEE-Mains Online-2021]**



(A) 1 Ω (B) 0 Ω (C) 3 Ω (D) $\frac{9}{2}\Omega$

18. Five identical cells each of internal resistance 1 Ω and emf 5 V are connected in series and in parallel with an external resistance 'R'. For what value of R, current in series and parallel combination will remain the same? **[JEE-Mains Online-2021]**

(A) 10 Ω (B) 1 Ω (C) 5 Ω (D) 25 Ω

19. What equal length of an iron wire and a copper-nickel alloy wire, each of 2 mm diameter connected parallel to give an equivalent resistance of 3 Ω ? **[JEE-Mains Online-2021]**

(Given resistivities of iron and copper-nickel alloy wire are 12 μΩ cm and 51 μΩ cm respectively)

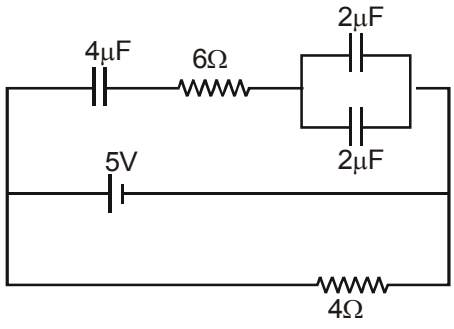
(A) 82 m (B) 110 m (C) 90 m (D) 97 m

20. If you are provided a set of resistance 2 Ω, 4 Ω, 6 Ω and 8 Ω. Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\Omega$. **[JEE-Mains Online-2021]**

(A) 6 Ω and 8 Ω are in parallel with 2 Ω and 4 Ω in series
(B) 2 Ω and 4 Ω are in parallel with 6 Ω and 8 Ω in series
(C) 2 Ω and 6 Ω are in parallel with 4 Ω and 8 Ω in series
(D) 4 Ω and 6 Ω are in parallel with 2 Ω and 8 Ω in series

21. Calculate the amount of charge on capacitor of 4 μF. The internal resistance of battery is

[JEE-Mains Online-2021]

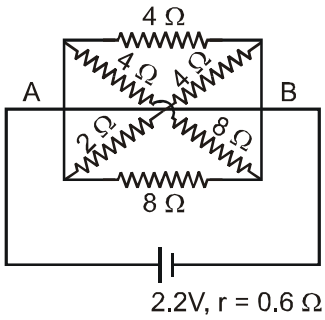


(A) 4 μC (B) zero (C) 8 μC (D) 16 μC

22. Two resistors $R_1 = (4 \pm 0.8)\ \Omega$ and $R_2 = (4 \pm 0.4)\ \Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be : **[JEE-Mains Online-2021]**

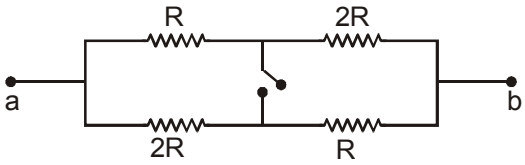
(A) $(2 \pm 0.3)\Omega$ (B) $(4 \pm 0.3)\ \Omega$ (C) $(2 \pm 0.4)\ \Omega$ (D) $(4 \pm 0.4)\ \Omega$

23. In the given figure, the emf of the cell is 2.2 V and if internal resistance is 0.6Ω. Calculate the power dissipated in the whole circuit : **[JEE-Mains Online-2021]**



(A) 1.32 W (B) 4.4 W (C) 0.65 W (D) 2.2 W

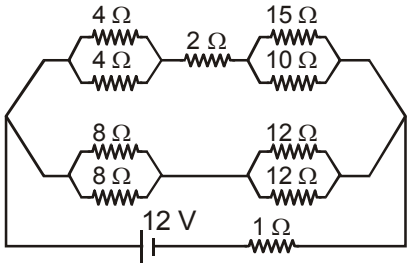
24. The ratio of the equivalent resistance of the netwrok (shown in figure) between the points a and b when switch is open and switch is closed is x : 8. The value of x is _____ . **[JEE-Mains Online-2021]**



25. First, a set of n equal resistors of 10 Ω each are connected in series to a battery of emf 20 V and internal resistance 10 Ω. A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is. **[JEE-Mains Online-2021]**

26. A resistor dissipates 192 J of energy in 1 s when a current of 4 A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5 s is _____ J. **[JEE-Mains Online-2021]**

27. The voltage drop across 15Ω resistance in the given figure will be _____ V. **[JEE-Mains Online-2021]**

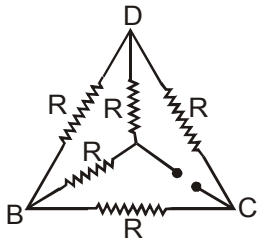


28. A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be _____ . **[JEE-Mains Online-2021]**

29. A current through a wire depends on time as $i = \alpha_0 t + \beta t^2$ **[JEE-Mains Online-2021]**
where $\alpha_0 = 20$ A/s and $\beta = 8$ As^{-2}. Find the charge crossed through a section of the wire in 15 s.
(A) 260 C (B) 2250 C (C) 11250 C (D) 2100 C

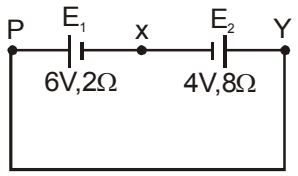
30. A wire of 1Ω has a lenght of 1m. It is streched till its length increases by 25%. The percentage change in resistance to the nearest integer is : **[JEE-Mains Online-2021]**
(A) 76% (B) 25% (C) 56% (D) 12.5%

31. Five equal resistances are connected in a network as shown in figure. The net resistance between the point A and B is : **[JEE-Mains Online-2021]**



(A) $\frac{R}{2}$ (B) R (C) $\frac{3R}{2}$ (D) 2R

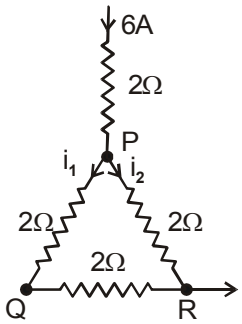
32. A cell E_1 of cmf 6V and internal resistance 2Ω is connected with another cell E_2 of emf 4V and internal resistance 8Ω (as shown in the figure). The potential difference across points X and Y is : **[JEE-Mains Online-2021]**



(A) 3.6 V (B) 2.0 V (C) 10.0 V (D) 5.6 V

33. A cylindrical wire of radius 0.5 mm and conductivity 5×10^7 S/m is subjected to an electric field of 10 mV/m. The expected value of current in the wire will be $x^3\pi$ mA. The value of x is_____. **[JEE-Mains Online-2021]**

34. A current of 6 A enters one comer P of an equilateral triangle PQR having 3 wires of resistance 2Ω each and leaves by the comer R. The currents i_1 in ampere is _____. **[JEE-Mains Online-2021]**



Question Stem for Question Nos. 35 and 36

In a circuit, a metal filament lamp is connected in series with a capacitor of capacitance C μF across a 200 V, 50 Hz supply. The power consumed by the lamp is 500 W while the voltage drop across it is 100 V. Assume that there is no inductive load in the circuit. Take rms values of the voltages. The magnitude of the phaseangle (in degrees) between the current and the supply voltage is φ. Assume, $\pi\sqrt{3} \approx 5$.

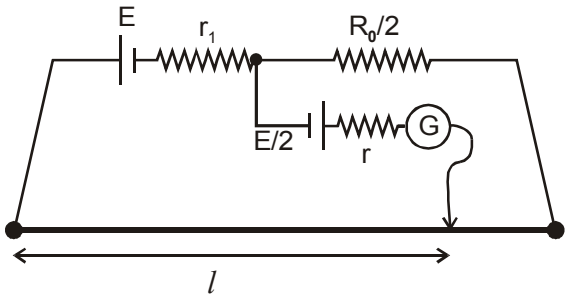
[IIT-JEE Advance 2021]

35. The value of C is ___.

36. The value of φ is ___.

37. In order to measure the internal resistance r_1 of a cell of emf E, a meter bridge of wire resistance $R_0 = 50\ \Omega$, a resistance $R_0/2$, another cell of emf E/2 (internal resistance r) and a galvanometer G are used in a circuit, as shown in the figure. If the null point is found at l = 72 cm, then the value of r_1 = ... Ω.

[IIT-JEE Advance 2021]



38. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Alloys such as constantan and manganin are used in making standard resistance coils.
Reason R : Constantan and manganin have very small value of temperature coefficient of resistance.
In the light of the above statements, choose the correct answer from the options given below.
(A) Both A and R are true and R is the correct explanation of A. **[JEE-Mains Online-2022]**
(B) Both A and R are true but R is NOT the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

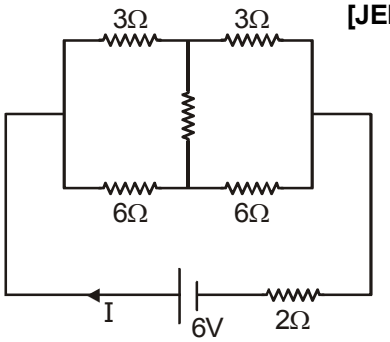
39. Two sources of equal emfs are connected in series. This combination is connected to an external resistance R. The internal resistances of the two sources are r_1 and r_2 ($r_1 > r_2$). If the potential difference across the source of internal resistance r_1 is zero then the value of R will be **[JEE-Mains Online-2022]**

(A) $r_1 - r_2$ (B) $\frac{r_1 r_2}{r_1 + r_2}$ (C) $\frac{r_1 + r_2}{2}$ (D) $r_2 - r_1$

40. A battery of 6V is connected to the circuit as shown below. The current I drawn from the battery is **[JEE-Mains Online-2022]**

(A) 1A (B) 2A

(C) $\frac{6}{11}$A (D) $\frac{4}{3}$A



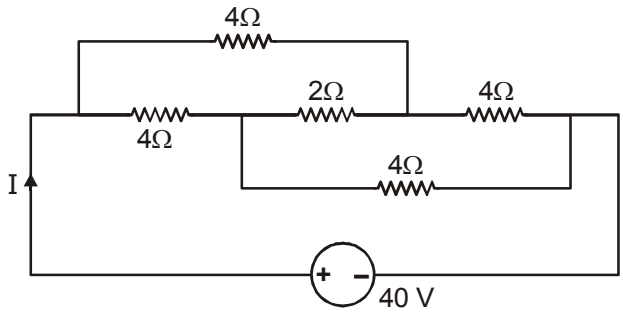
41. (A) The drift velocity of electrons decreases with the increase in the temperature of conductor.
(B) The drift velocity is inversely proportional to the area of cross-section of given conductor.
(C) The drift velocity does not depend on the applied potential difference to the conductor.
(D) The drift velocity of electron is inversely proportional to the length of the conductor.
(E) The drift velocity increase with the increase in the temperature of conductor.
Choose the correct answer from the option given below **[JEE-Mains Online-2022]**
(A) A and B only (B) A and D only (C) B and E only (D) B and C only

42. Given below are two statements :
Statement I : A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω.
Statement II : Two resistance 2R and 3R are connected in parallel in a electric circuit. The value of thermal energy developed in 3R and 2R will be in the ratio 3 : 2.
In the light of the above statements, choose the most appropriate answer from the options given below
(A) Both statement I and statement II are correct **[JEE-Mains Online-2022]**
(B) Both statement I and statement II are incorrect
(C) Statement I is correct but statement II is incorrect
(D) Statement I is incorrect but statement II is correct.

43. Two metallic wires of identical dimensions are connected is series. If σ_1 and σ_2 are the conductivities of the these wires respectively, the effective conductivity of the combination is : **[JEE-Mains Online-2022]**

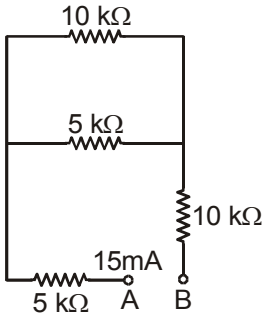
(A) $\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (B) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (C) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$ (D) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

44. The current I in the given circuit will be **[JEE-Mains Online-2022]**



(A) 10 A (B) 20 A (C) 4 A (D) 40 A

45. A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be **[JEE-Mains Online-2022]**



(A) 50 V (B) 75 V (C) 150 V (D) 275 V

46. A 1 m long wire is broken into two unequal parts X and Y The X part of the wire is streched into another wire W. Length of W is twice the length of X and the resistance of W is twice that of Y. Find the ratio of length of X and Y. **[JEE-Mains Online-2022]**

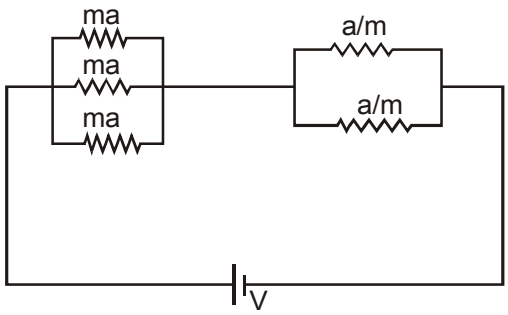
(A) 1 : 4 (B) 1 : 2 (C) 4 :1 (D) 2 : 1

47. A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is : **[JEE-Mains Online-2022]**

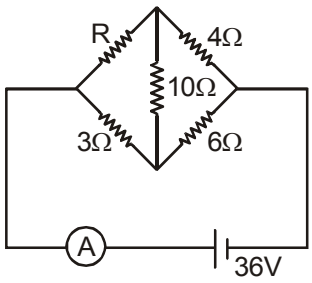
(A) 9 : 1 (B) 1 : 9 (C) 4 : 1 (D) 3 : 1

48. The current density in a cylindrical wire of radius r = 4.0 mm is 1.0×10^6 A/m^2. The current through the outer portion of the wire between radial distances r/2 and r is $x\pi$ A; where x is_____. **[JEE-Mains Online-2022]**

49. In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be $\sqrt{\frac{x}{2}}$. The value of x is _____ . **[JEE-Mains Online-2022]**



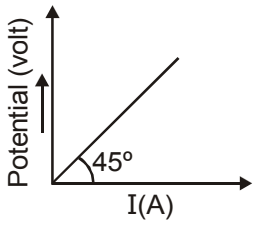
50. Current measured by the ammeter (A) in the reported circuit when no current flows through 10 Ω resistance will be _____ A. **[JEE-Mains Online-2022]**



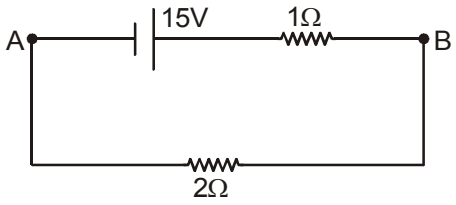
51. Two coils require 20 minutes and 60 minutes respectively to produce same amount of heat energy when connected separately to the same source. If they are connected in parallel arrangement to the same source; the time required to produce same amount of heat by the combination of coils, will be_____min. **[JEE-Mains Online-2022]**

52. The length of a given cylindrical wire is increased to double of its original length. The persentage increase in the resistance of the wire will be _____%. **[JEE-Mains Online-2022]**

53. The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is 31.4 cm. The diameter of wire is measured as 2.4 cm. The resistivity of the given wire is measured as $x \times 10^{-3}\ \Omega$ cm. The value of x is _____. [Take $\pi = 3.14$] **[JEE-Mains Online-2022]**

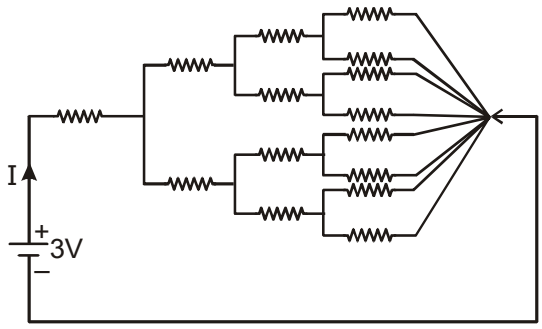


54. For the network shown below, the value $V_B - V_A$ is _____V. **[JEE-Mains Online-2022]**



55. All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5}$ A. The value of a is _____.

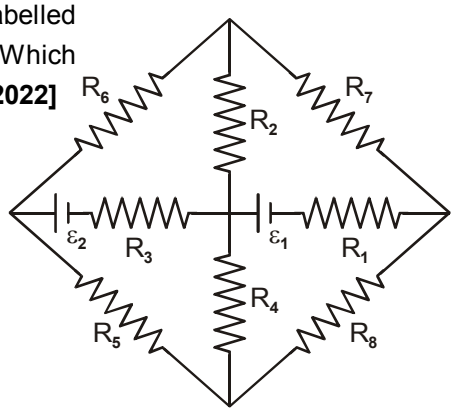
[JEE-Mains Online-2022]



56. The current density in a cylindrical wire of radius 4 mm is $4 \times 10^6\ Am^{-2}$. The current through the outer portion of the wire between radial distances $\frac{R}{2}$ and R is _____ π A. **[JEE-Mains Online-2022]**

57. A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed thorugh it. If the current increases to 3 A, the energy developed in 10 s is _____J. **[JEE-Mains Online-2022]**

58. The figure shows a circuit having eight resistances of 1Ω each, labelled R_1 to R_8, and two ideal batteries with voltages ε_1 = 12V & ε_2 = 6V. Which of the following statement(s) is(are) correct? **[IIT-JEE Advance 2022]**



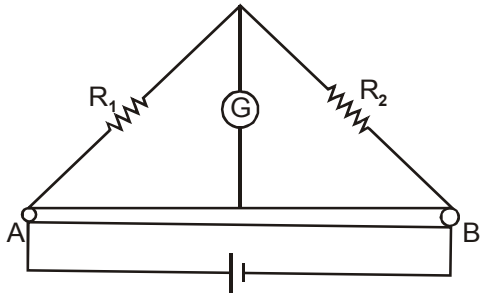
(A) The magnitude of current flowing through R_1 is 7.2 A.

(B) The magnitude of current flowing through R_2 is 1.2 A.

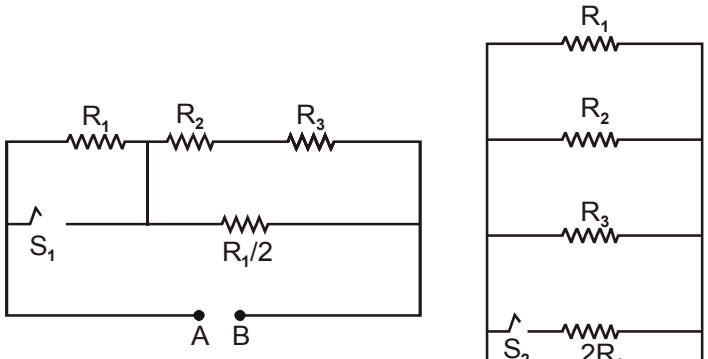
(C) The magnitude of current flowing through R_3 is 4.8 A.

(D) The magnitude of current flowing through R_5 is 2.4 A.

59. Two resistances R_1 = X Ω and R_2 = 1 Ω are connected to a wire AB of uniform resistivity, as shown in the figure. The radius of the wire varies linearly along its axis from 0.2 mm at A to 1 mm at B. A galvanometer (G) connected to the center of the wire, 50 cm from each end along its axis, shows zero deflection when A and B are connected to a battery. The value of X is _____. **[IIT-JEE Advance 2022]**



60. In Circuit-1 and Circuit-2 shown in the figures, R_1 = 1 Ω, R_2 = 2 Ω and R_3 = 3 Ω .
P_1 and P_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in open conditions, respectively.
Q_1 and Q_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in closed conditions, respectively. **[IIT-JEE Advance 2022]**



Which of the following statement(s) is(are) correct?

(A) When a voltage source of 6V is connected across A and B in both circuits, $P_1 < P_2$

(B) When a constant current source of 2 Amp is connected across A and B in both circuits, $P_1 > P_2$.

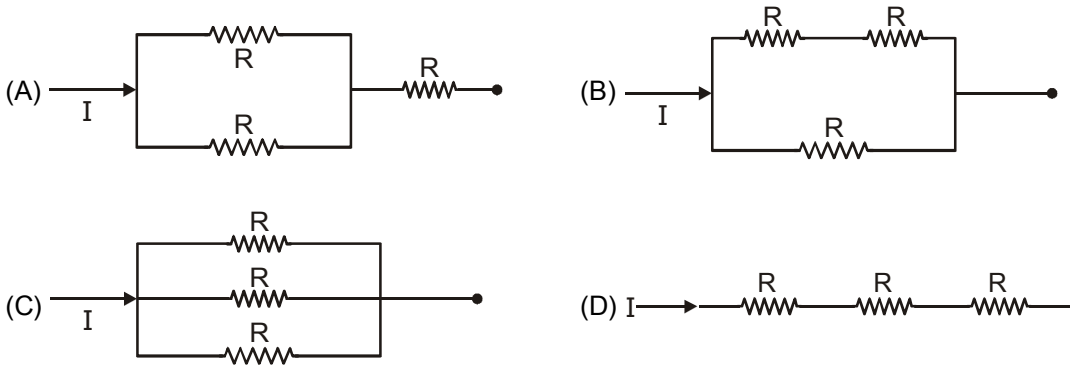
(C) When a voltage source of 6 V is connected across A and B in Circuit-1, $Q_1 > P_1$

(D) When a constant current source of 2 Amp is connected across A and B in both circuits, $Q_2 < Q_1$.

61. In a metallic conductor, under the effect of applied electric field, the free electrons of the conductor
(A) move in the straight line path in the same direction **[JEE-Mains Online-2023]**
(B) drift from higher potential to lower potential.
(C) move in the curved paths from lower potential to higher potential
(D) move with the uniform velocity throughout from lower potential to higher potential

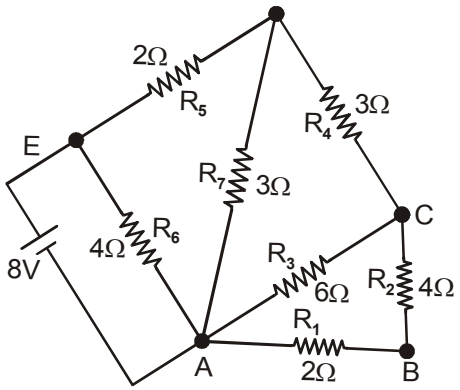
62. Given below are two statements :
Statement-I : The equivalent resistance of resistors in a series combination is smaller than least resistance used in the combination. **[JEE-Mains Online-2023]**
Statement-II : The resistivity of the material is independent of temperature.
In the light of the above statements, choose the correct answer from the options given below :
(A) Statement I is false but Statement II is true (B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false (D) Both Statement I and Statement II are true

63. Different combination of 3 resistors of equal resistance R are shown in the figures. The increasing order for power dissipation is : **[JEE-Mains Online-2023]**



(A) $P_B < P_C < P_D < P_A$ (B) $P_A < P_B < P_C < P_D$
(C) $P_C < P_D < P_A < P_B$ (D) $P_C < P_B < P_A < P_D$

64. The current flowing through R_2 is **[JEE-Mains Online-2023]**



(A) $\frac{1}{3}$A (B) $\frac{1}{2}$A (C) $\frac{2}{3}$A (D) $\frac{1}{4}$A

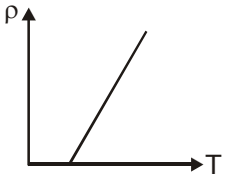
65. A wire of resistance 160 Ω is melted and drawn in a wire of one-fourth of its length. The new resistance of the wire will be **[JEE-Mains Online-2023]**

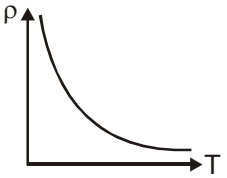
(A) 10 Ω (B) 16 Ω (C) 640 Ω (D) 40 Ω

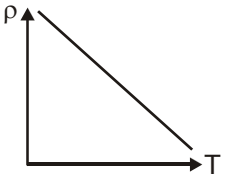
66. Two identical heater filaments are connected first in parallel and then in series. At the same applied voltage, the ratio of heat produced in same time for parallel to series will be : **[JEE-Mains Online-2023]**

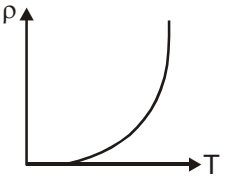
(A) 1 : 2 (B) 1 : 4 (C) 4 : 1 (D) 2 : 1

67. The resistivity (ρ) of semiconductor varies with temperature. Which of the following curve represents the correct behaviour **[JEE-Mains Online-2023]**

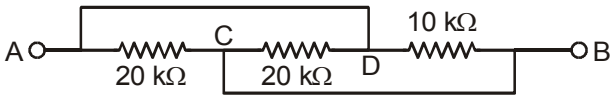
(A) 

(B) 

(C) 

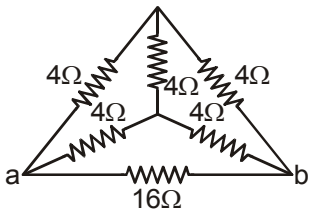
(D) 

68. The equivalent resistance between A and B as shown in figure is : **[JEE-Mains Online-2023]**



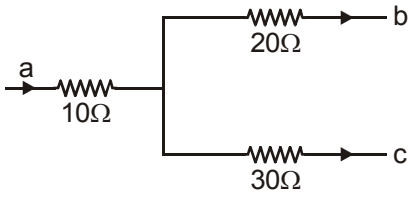
(A) 5 kΩ (B) 20 kΩ (C) 10 kΩ (D) 30 kΩ

69. The equivalent resistance of the circuit shown below between points a and b is **[JEE-Mains Online-2023]**



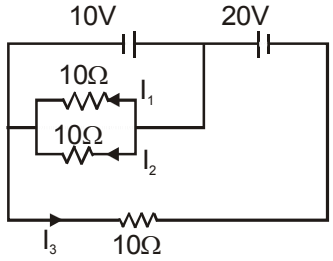
(A) 3.2 Ω (B) 24 Ω (C) 20 Ω (D) 16 Ω

70. Figure shows a part of an electric circuit. The potentials at points a, b and c are 30 V, 12 V and 2 V respectively. The current through the 20Ω resistor will be **[JEE-Mains Online-2023]**



(A) 0.4 A (B) 1.0 A (C) 0.6 A (D) 0.2 A

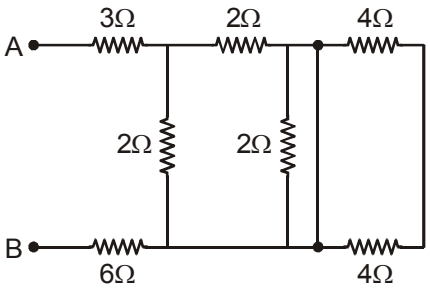
71. In the given circuit, the value of $\left|\frac{I_1 + I_3}{I_2}\right|$ is _____ **[JEE-Mains Online-2023]**



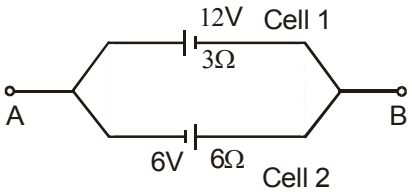
72. A hollow cylindrical conductor has length of 3.14 m, while its inner and outer diameters are 4 mm and 8 mm respectively. The resistance of the conductor is n × 10^{-3} Ω. If the resitivity of the material is 2.4 × 10^{-8} Ωm. The value of n is _____. **[JEE-Mains Online-2023]**

73. If a copper wire is stretched to increase its length by 20%. The percentage increase in resistance of the wire is _____%. **[JEE-Mains Online-2023]**

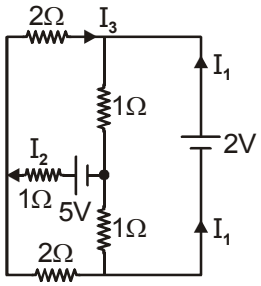
74. In the given circuit, the equivalent resistance between the terminal A and B is _____Ω. **[JEE-Mains Online-2023]**



75. Two cells are connected between points A and B as shown. Cell 1 has emf of 12 V and internal resistance of 3Ω. Cell 2 has emf of 6V and internal resistance of 6Ω. An external resistor R of 4Ω is connected across A and B. The current flowing through R will be _____ A. **[JEE-Mains Online-2023]**

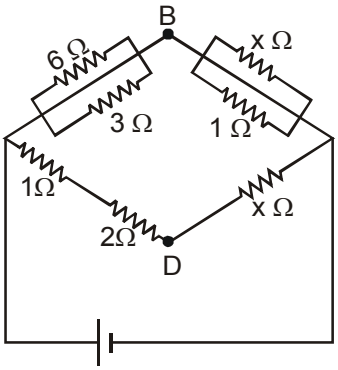


76. In the following circuit, the magnitude of current I_1, is_____A. **[JEE-Mains Online-2023]**



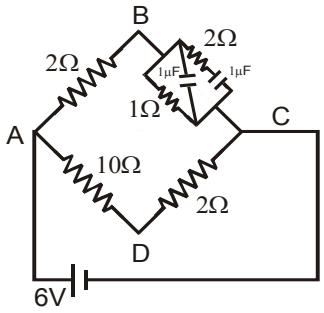
77. If the potential difference between B and D is zero, the value of x is $\frac{1}{n}\Omega$. The value of n is _____.

[JEE-Mains Online-2023]



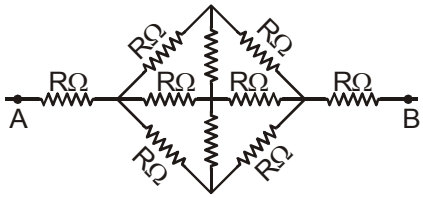
78. Two identical cells, when connected either in parallel or in series gives same current in an external resistance 5Ω. The internal resistance of each cell will be _____ Ω. **[JEE-Mains Online-2023]**

79. For the given circuit, in the steady state, $|V_B - V_D|$ = _____ V. **[JEE-Mains Online-2023]**



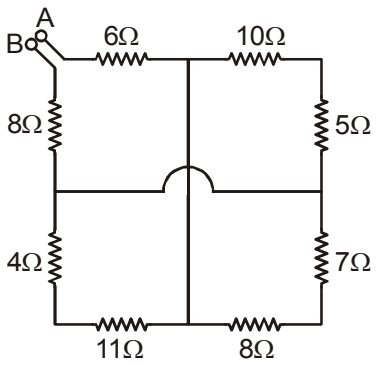
80. The effecitive resistance between A and B, if resistance of each resistor is R, will be :

[JEE-Mains Online-2024]



(A) $\frac{2}{3}R$ (B) $\frac{8R}{3}$ (C) $\frac{5R}{3}$ (D) $\frac{4R}{3}$

81. The equivalent resistance between A and B is **[JEE-Mains Online-2024]**



(A) 18Ω (B) 27Ω (C) 19Ω (D) 25Ω

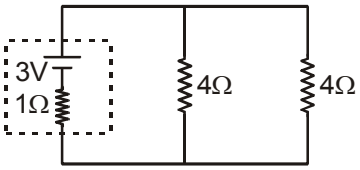
82. A galvanmeter has a coil ofresistance 200 Ω with a full scale deflection at 20 μA. The value of resistance to be added to use its as an ammeter of range (0 – 20) mA is : **[JEE-Mains Online-2024]**

(A) 0.40 Ω (B) 0.20 Ω (C) 0.50 Ω (D) 0.10 Ω

83. Water boils in an electric kettle in 20 minutes after being switched on. Using the same main supply, the length of the heating element should be _____ to _____ times of its initial length if the water is to be boiled in 15 minutes. **[JEE-Mains Online-2024]**

(A) decreased, 3/4 (B) increased, 4/3 (C) decreased, 4/3 (D) increased, 3/4

84. In the given circuit, the terminal potential difference of the cell is : **[JEE-Mains Online-2024]**

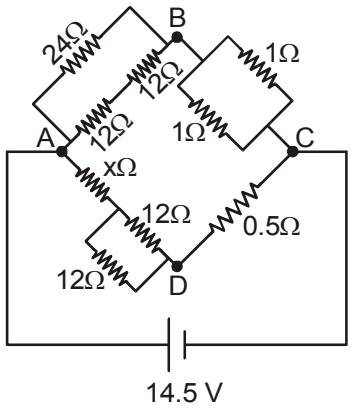


(A) 3 V (B) 2 V (C) 1.5 V (D) 4 V

85. The number of electrons flowing per second in the filament of a 110 W bulb operating at 220 V is : (Given e = 1.6×10^{-19} C) **[JEE-Mains Online-2024]**

(A) 6.25×10^{18} (B) 6.25×10^{17} (C) 1.25×10^{19} (D) 31.25×10^{17}

86. The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is **[JEE-Mains Online-2024]**

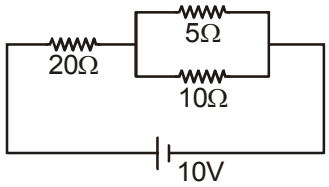


(A) 42 Ω (B) 3 Ω (C) 9 Ω (D) 6 Ω

87. To measure the internal resistance of a battery, potentiometer is used. For R = 10 Ω, the balance point is observed at ℓ = 500 cm and for R = 1 Ω the balance point is observed at ℓ = 400 cm. The internal resistance of the battery is approximately **[JEE-Mains Online-2024]**

(A) 0.2 Ω (B) 0.3 Ω (C) 0.4 Ω (D) 0.1 Ω

88. The ratio of heat dissipated per second through the resistance 5 Ω and 10 Ω in the circuit given below is : **[JEE-Mains Online-2024]**



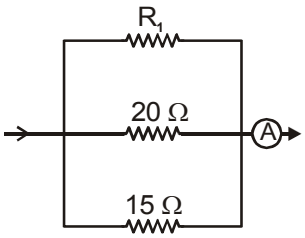
(A) 4 : 1 (B) 2 : 1 (C) 1 : 1 (D) 1 : 2

89. An electric bulb rated 50 W - 200 V is connected across a 100 V supply. The power dissipation of the bulb is : **[JEE-Mains Online-2024]**

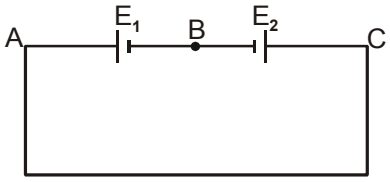
(A) 25 W (B) 12.5 W (C) 50 W (D) 100 W

90. The current in a conductor is expressed as $I = 3t^2 + 4t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during t = 1s to t = 2s is _____C. **[JEE-Mains Online-2024]**

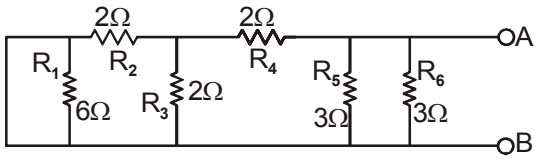
91. In the given citcuit, the current flowing through the resistance 20 Ω is 0.3 A, while the ammeter reads 0.9 A. The value of R_1 is _____ Ω. **[JEE-Mains Online-2024]**



92. Two cells are connected in opposition as shown. Cell E_1 is of 8 V emf and 2 Ω internal resistance; the cell E_2 is of 2 V emf and 4Ω internal resistance. The terminal potential difference of cell E_2 is _____V. **[JEE-Mains Online-2024]**

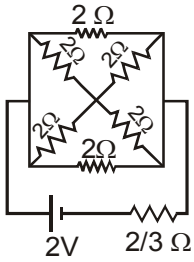


93. Equivalent resistance of the following network is _____Ω. **[JEE-Mains Online-2024]**



94. Two resistance of 100 Ω and 200 Ω are connected in series with a battery of 4 V and negligible internal resistance. A voltmeter is used to measure voltage across 100 Ω resistance, which gives reading as 1 V. The resistance of voltmeter must be _____ Ω. **[JEE-Mains Online-2024]**

95. In the following circuit, the battery has an emf of 2V and an internal resistance of $\frac{2}{3}\Omega$. The power consumption in the entire circuit is _____W. **[JEE-Mains Online-2024]**



96. Match **List-I** with **List-II** **[JEE-Mains Online-2025]**

	List-I		**List-II**
(A)	Angular Impulse	(I)	$[M^0 L^2 T^{-2}]$
(B)	Latent Heat	(II)	$[M L^2 T^{-3} A^{-1}]$
(C)	Electrical resistivity	(III)	$[M L^2 T^{-1}]$
(D)	Electromotive force	(IV)	$[M L^3 T^{-3} A^{-2}]$

Choose the correct answer from the options given below :

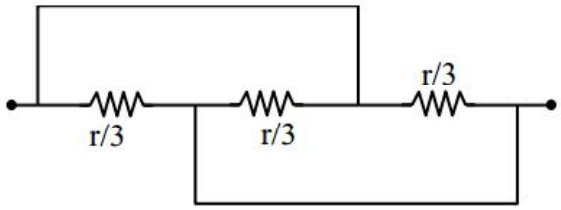
(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) (B) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(C) (A)-(III), (B)-(I), (C)-(II), (D)-(IV) (D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

97. A wire of resistance R is bent into an equilateral triangle and an identical wire is bent into a square. The ratio of resistance between the two end points of an edge of the triangle to that of the square is : **[JEE-Mains Online-2025]**

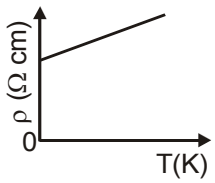
(A) 9/8 (B) 8/9 (C) 27/32 (D) 32/27

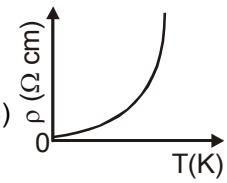
98. Find the equivalent resistance between two ends of the following circuit. **[JEE-Mains Online-2025]**

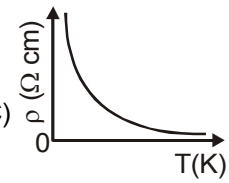


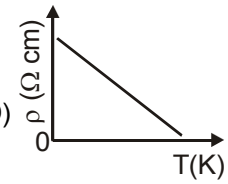
(A) r (B) $\frac{r}{6}$ (C) $\frac{r}{9}$ (D) $\frac{r}{3}$

99. Which of the following resistivity (ρ) v/s temperature (T) curves is most suitable to be used in wire bound standard resistors? **[JEE-Mains Online-2025]**

(A) 

(B) 

(C) 

(D) 

100. Given below are two statements : **[JEE-Mains Online-2025]**

Statement-I : The equivalent emf of two nonideal batteries connected in parallel is smaller than either of the two emfs.

Statement-II : The equivalent internal resistance of two nonideal batteries connected in parallel is smaller than the internal resistance of either of the two batteries.

In the light of the above statements, choose the **correct** answer from the options given below.

(A) Statement-I is true but Statement-II is false (B) Both Statement-I and Statement-II are false

(C) Both Statement-I and Statement-II are true (D) Statement-I is false but Statement-II is true

101. The battery of a mobile phone is rated as 4.2 V, 5800 mAh. How much energy is stored in it when fully charged? **[JEE-Mains Online-2025]**

(A) 43.8 kJ (B) 48.7 kJ (C) 87.7 kJ (D) 24.4 kJ

102. A motor operating on 100 V draws a current of 1 A. If the efficiency of the motor is 91.6%, then the loss of power in units of cal/s is **[JEE-Mains Online-2025]**

(A) 4 (B) 8.4 (C) 2 (D) 6.2

103. A wire of length 25 m and cross-sectional area 5 mm2 having resistivity of 2 × 10–6 ? m is bent into a complete circle. The resistance between diametrically opposite points will be **[JEE-Mains Online-2025]**

(A) 12.5 Ω (B) 50 Ω (C) 100 Ω (D) 25 Ω

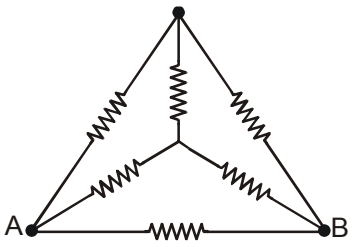
104. There are 'n' number of identical electric bulbs, each is designed to draw a power p independently from the mains supply. They are now joined in series across the main supply. The total power drawn by the combination is **[JEE-Mains Online-2025]**

(A) np (B) $\frac{p}{n^2}$ (C) $\frac{p}{n}$ (D) p

105. Current passing through a wire as function of time is given as I(t) = 0.02 t + 0.01 A. The charge that will flow through the wire from t = 1s to t = 2s is : **[JEE-Mains Online-2025]**

(A) 0.06 C (B) 0.02 C (C) 0.07 C (D) 0.04 C

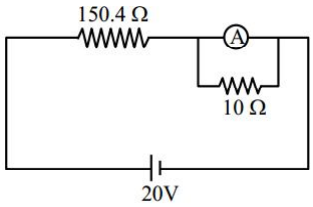
106. A wire of resistance R is bent into a triangular pyramid as shown in figure with each segment having same length. The resistance between points A and B is R/n. The value of n is **[JEE-Mains Online-2025]**



(A) 16 (B) 14 (C) 10 (D) 12

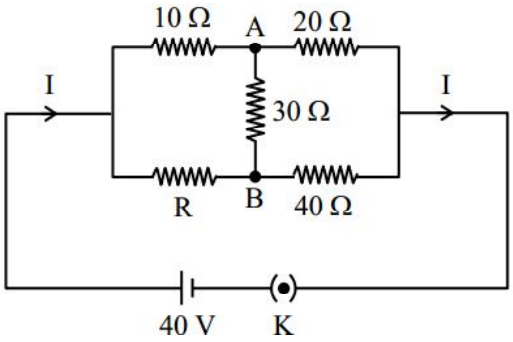
107. Two cells of emf 1V and 2V and internal resistance 2 Ω and 1 Ω, respectively, are connected in series with an external resistance of 6 Ω. The total current in the circuit is I_1. Now the same two cells in parallel configuration are connected to same external resistance. In this case, the total current drawn is I_2. The value of $\left(\frac{I_1}{I_2}\right)$ is $\frac{x}{3}$. The value of x is _____. **[JEE-Mains Online-2025]**

108. In the figure shown below, a resistance of 150.4 Ω is connected in series to an ammeter A of resistance 240 Ω. A shunt resistance of 10 Ω is connected in parallel with the ammeter. The reading of the ammeter is_____ mA. **[JEE-Mains Online-2025]**



109. A wire of resistance 9 Ω is bent to form an equilateral triangle. Then the equivalent resistance across any two vertices will be _____ ohm. **[JEE-Mains Online-2025]**

110. The value of current I in the electrical circuit as given below, when potential at A is equal to the potential at B, will be _____ A. **[JEE-Mains Online-2025]**



111. A temperature difference can generate e.m.f. in some materials. Let S be the e.m.f. produced per unit temperature difference between the ends of a wire, σ the electrical conductivity and κ the thermal conductivity of the material of the wire. Taking M, L, T, I and K as dimensions of mass, length, time, current and temperature, respectively, the dimensional formula of the quantity $Z = \frac{S^2\sigma}{\kappa}$ is : **[IIT-JEE Advance 2025]**

(A) $[M^0L^0T^0I^0K^0]$ (B) $[M^0L^0T^0I^0K^{-1}]$ (C) $[M^1L^2T^{-2}I^{-1}K^{-1}]$ (D) $[M^1L^2T^{-4}I^{-1}K^{-1}]$

ANSWER KEY

EXERCISE-I

ONLY ONE OPTION IS CORRECT

1.	C	**2.**	C	**3.**	D	**4.**	C	**5.**	B	**6.**	B	**7.**	D
8.	A	**9.**	C	**10.**	B	**11.**	A	**12.**	C	**13.**	C	**14.**	D
15.	C	**16.**	C	**17.**	B	**18.**	B	**19.**	D	**20.**	B	**21.**	C
22.	B	**23.**	B	**24.**	B	**25.**	A	**26.**	C	**27.**	B	**28.**	D
29.	A	**30.**	A	**31.**	D	**32.**	D	**33.**	D	**34.**	A	**35.**	A
36.	B	**37.**	D	**38.**	C	**39.**	D	**40.**	D	**41.**	B	**42.**	C
43.	A	**44.**	A	**45.**	B	**46.**	B	**47.**	B	**48.**	D	**49.**	D
50.	A	**51.**	B	**52.**	D	**53.**	D	**54.**	A	**55.**	B	**56.**	B
57.	C	**58.**	C	**59.**	A	**60.**	C	**61.**	A	**62.**	D	**63.**	C
64.	C	**65.**	B	**66.**	B								

ONE OR MORE THAN ONE OPTION MAY BE CORRECT

67.	ABCD	**68.**	D	**69.**	AD	**70.**	C	**71.**	A	**72.**	ABC	**73.**	BC
74.	AC	**75.**	A	**76.**	D	**77.**	AD	**78.**	C	**79.**	BC	**80.**	ABC
81.	A	**82.**	AB	**83.**	B	**84.**	B	**85.**	B	**86.**	D	**87.**	D
88.	B	**89.**	A	**90.**	B								

EXERCISE-II

1. (a) $J_0A/3$; (b) $2J_0A/3$ **2.** $\frac{R_2}{R_1}=\frac{3}{5}$ **3.** 4Ω **4.** 600Ω

5. 60 ohms **6.** $5\ \Omega$ **7.** 1 V **8.** I = 2.5 A, V = 3.5 Volts

9. $\sqrt{R_1R_2}$ **10.** 12A, –20W **11.** $\frac{22}{35}\Omega$ **12.** 1Ω

13. $\frac{3r}{5}$ **14.** (i) 10.52Ω ; (ii) 0.3125Ω

15. (i) $\frac{(k-1)^2}{k}$; $\frac{k}{(k-1)}$ (ii) $\frac{\left((k-1)/k^2\right)V_0}{R_3}$ **16.** 4/9 kg/sec., 450 sec

17. (i) V_{ab} = – 12 V, (ii) 3 amp from b to a **18.** 7.2 m **19.** $-\frac{22}{9}$V

20. 46.67 cm **21.** $\frac{10}{3}\ \Omega$, $5\ \Omega$ **22.** 7.5 m, 8.75 m, 6.25 m

23. (a) 1.01 Ω, (b) 0-5A, 0-10V, (c) 0.05 A

EXERCISE-III

1.	D	**2.**	C	**3.**	C	**4.**	4.00	**5.**	5.00
6.	C	**7.**	ABCD	**8.**	A, B, D	**9.**	C	**10.**	5.00
11.	B	**12.**	1.00	**13.**	B, C	**14.**	C, D		
15.	0.26 to 0.28	**16.**	A	**17.**	A	**18.**	B	**19.**	D
20.	B	**21.**	C	**22.**	A	**23.**	D	**24.**	09.00
25.	20.00	**26.**	3840	**27.**	06.00	**28.**	03.00	**29.**	C
30.	C	**31.**	B	**32.**	D	**33.**	5.00	**34.**	2.00
35.	90.00 - 105.00	**36.**	55.00 - 62.00	**37.**	3.00	**38.**	A	**39.**	A
40.	A	**41.**	B	**42.**	C	**43.**	B	**44.**	A
45.	D	**46.**	B	**47.**	A	**48.**	12.00	**49.**	3.00
50.	10.00	**51.**	15.00	**52.**	300	**53.**	144	**54.**	10.00
55.	8.00	**56.**	48	**57.**	450	**58.**	A, B, C, D	**59.**	5.00
60.	A,B,C	**61.**	C	**62.**	B	**63.**	D	**64.**	A
65.	A	**66.**	C	**67.**	B	**68.**	A	**69.**	A
70.	A	**71.**	2.00	**72.**	2.00	**73.**	44.00	**74.**	10.00
75.	1.00	**76.**	2.00	**77.**	2.00	**78.**	5.00	**79.**	1.00
80.	B	**81.**	C	**82.**	B	**83.**	A	**84.**	B
85.	D	**86.**	D	**87.**	B	**88.**	B	**89.**	B
90.	22.00	**91.**	30.00	**92.**	6.00	**93.**	1.00	**94.**	200
95.	3.00	**96.**	A	**97.**	D	**98.**	C	**99.**	A
100.	D	**101.**	C	**102.**	C	**103.**	D	**104.**	C
105.	D	**106.**	D	**107.**	4.00	**108.**	125	**109.**	2.00
110.	2.00	**111.**	B						